

2110403 - Introduction to Data Science and Data Engineering

Data Extraction

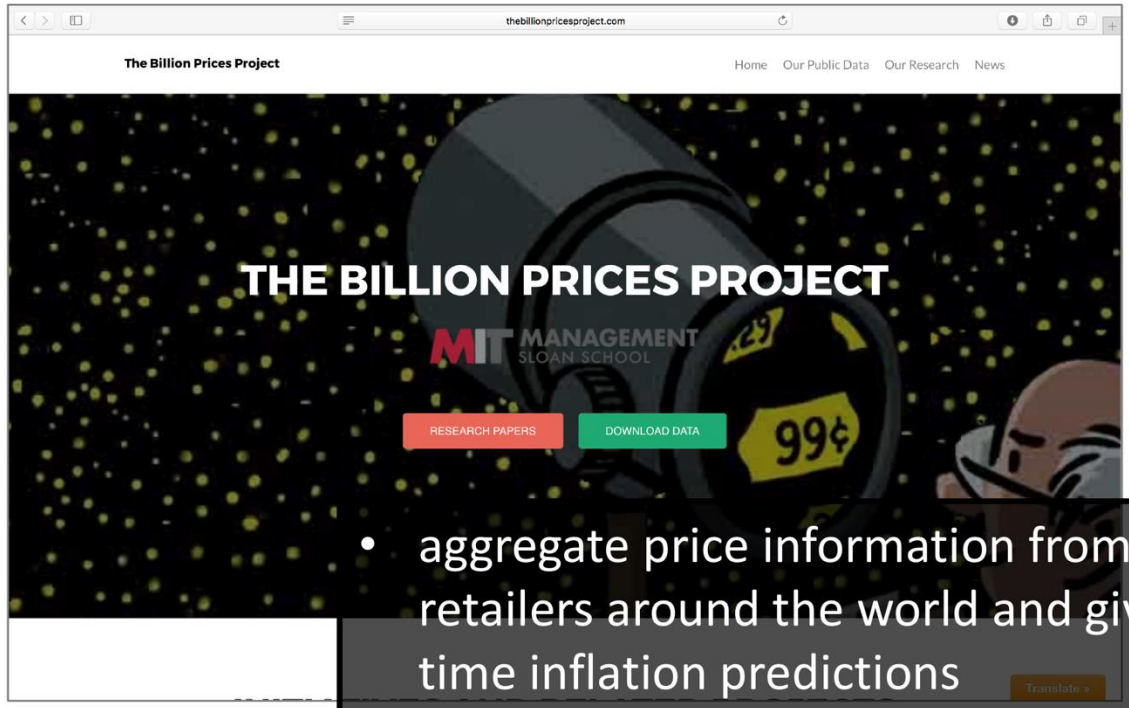
Asst.Prof. Natawut Nupairoj, Ph.D.
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Chulalongkorn University
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Internet Data Sources

- Internet provides lots of data sources
  - Financial
  - Weather
  - Sports
  - News
- Google creates business by utilizing Internet information
- Most data sources are human-readable, it is also possible to extract information systematically with program

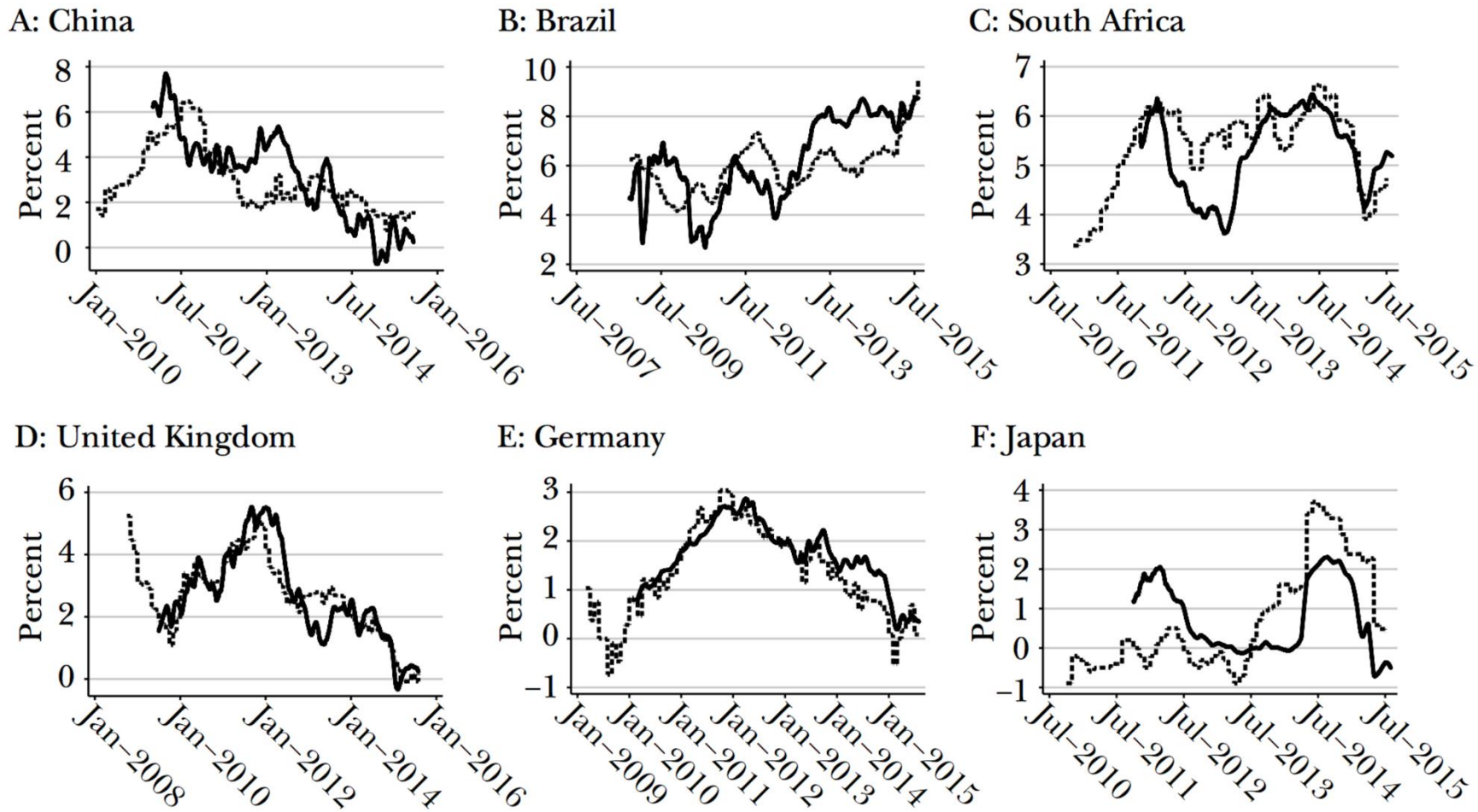
MIT’s Billion Prices Project



- aggregate price information from multitude of online retailers around the world and gives real time inflation predictions
- monitor daily price fluctuations of ~5 million items sold by ~300 online retailers in more than 70 countries

Cavallo, A., & Rigobon, R. (2016). The billion prices project: Using online prices for measurement and research. *The Journal of Economic Perspectives*, *30*(2), 151-178.

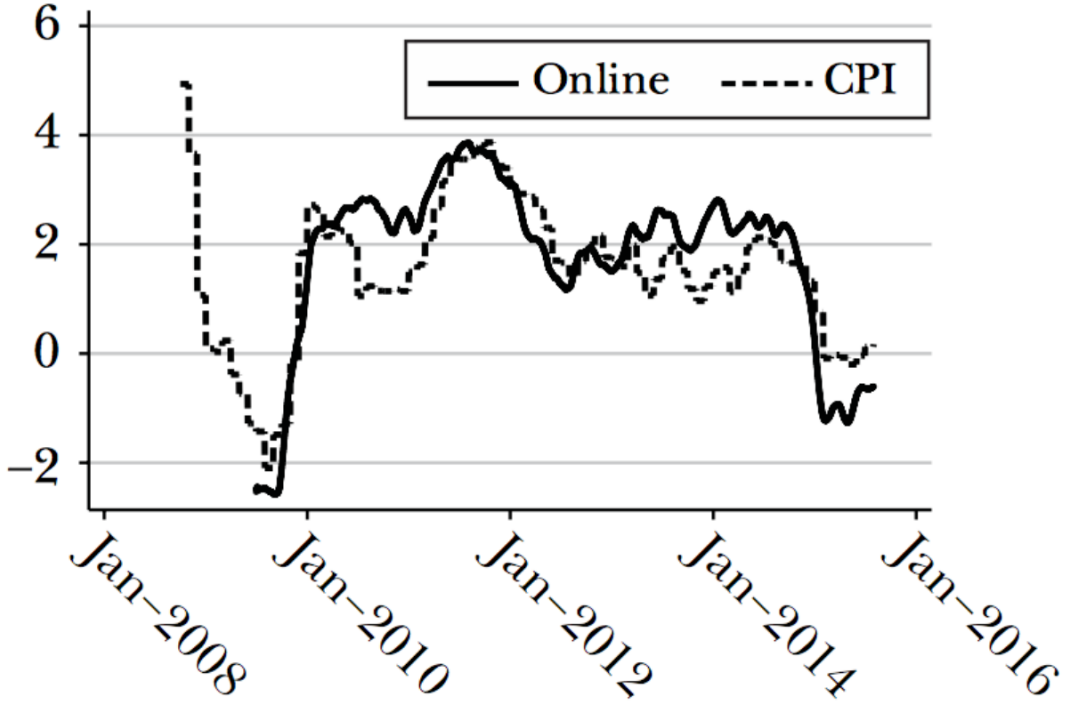
Annual Inflation Rate: Online vs. CPI



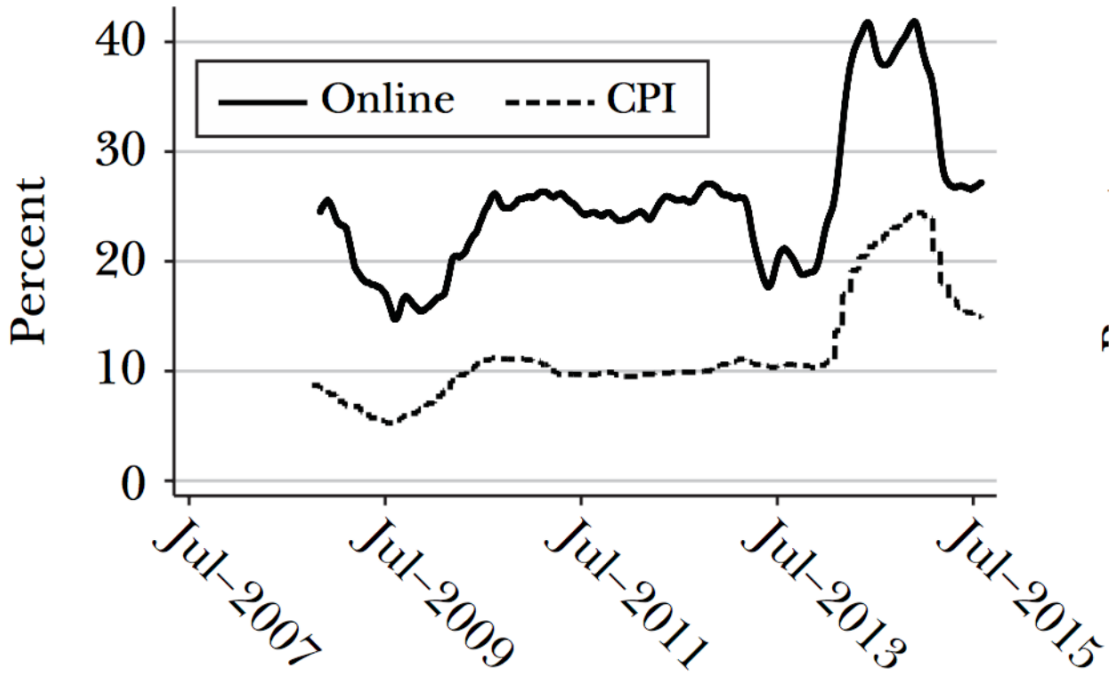
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Annual Inflation Rate: Online vs. CPI

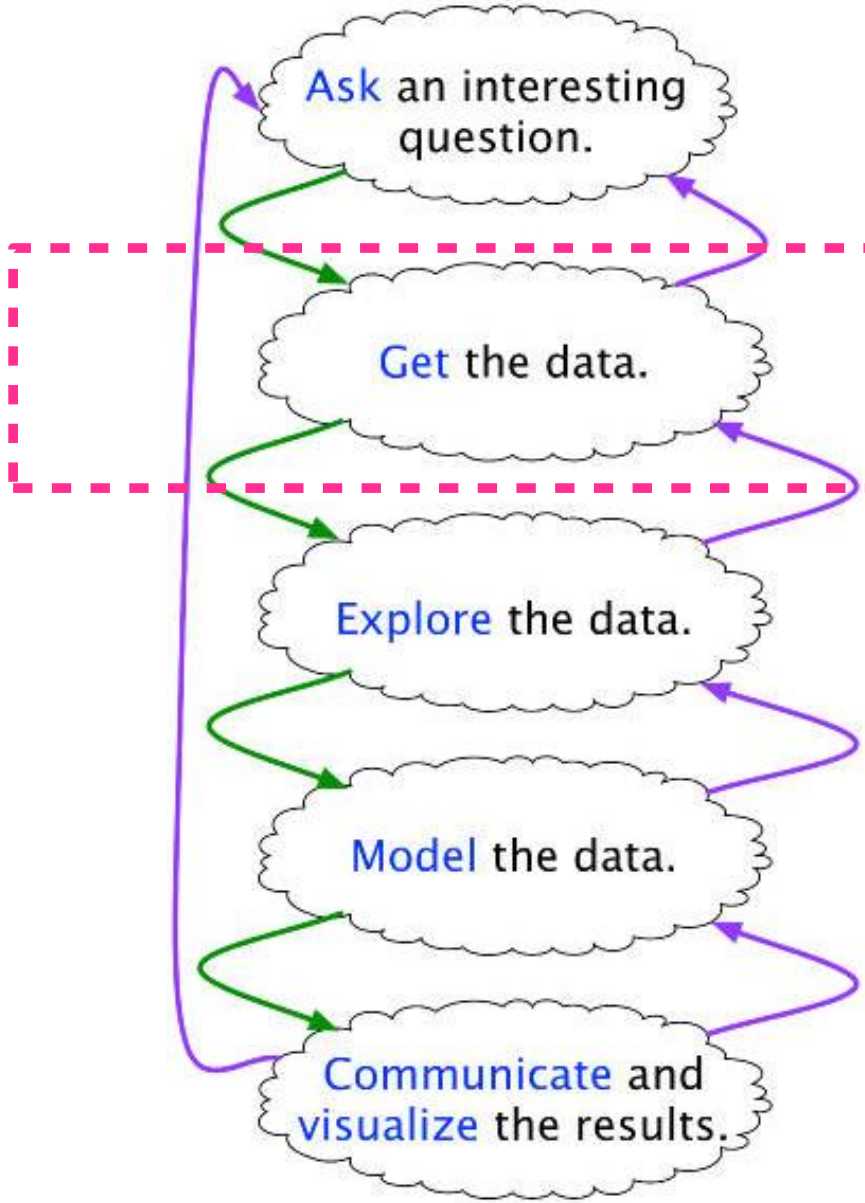
USA



Argentina



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Scrapy
Requests
BeautifulSoup
Selenium

pandas
statsmodels
NLTK
matplotlib
seaborn
folium
Spark

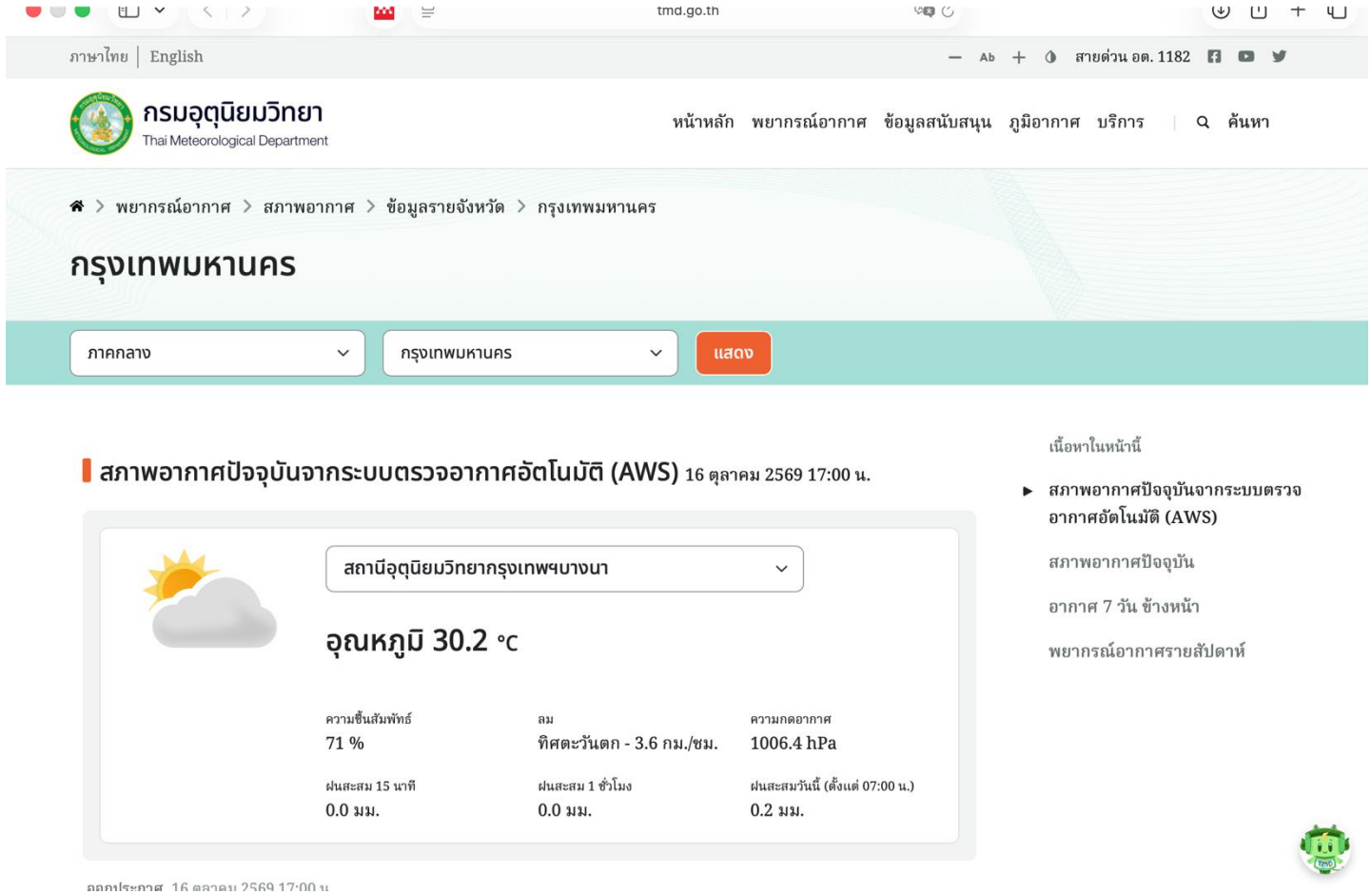
scikit-learn
Theano
Tensorflow
SparkML

Seaborn
Bokeh
Basemap
NetworkX

Data Extraction

- Data extraction is the process of getting data from a source for further data processing, storage or analysis elsewhere
- Many data sources are in unstructured or human-readable format
  - Gather organization structure from company's website
  - Extract accounting information from PDF financial reports
  - Collect relationship information from social media platform

Example Data Source: Thai Meteorological Department



Source: https://www.tmd.go.th/weatherForecast7Days?province=กรุงเทพมหานคร&culture=th-TH

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Data Extraction Steps

- Data Accessing
  - Reading a file
  - Issue a HTTP request
  - Call REST API using Request
  - Call Third-Party API using compatible library
  - Collect complicated web page with Selenium

Data Extraction Steps

- Data Parsing
  - Parsing HTML page using BeautifulSoup
  - Parsing HTML page using GenAI (Simple but Expensive)
  - Parsing text using regular expression
  - Parsing docx using python-docx
  - Parsing excel using Pandas
  - Parsing JSON string using JSON

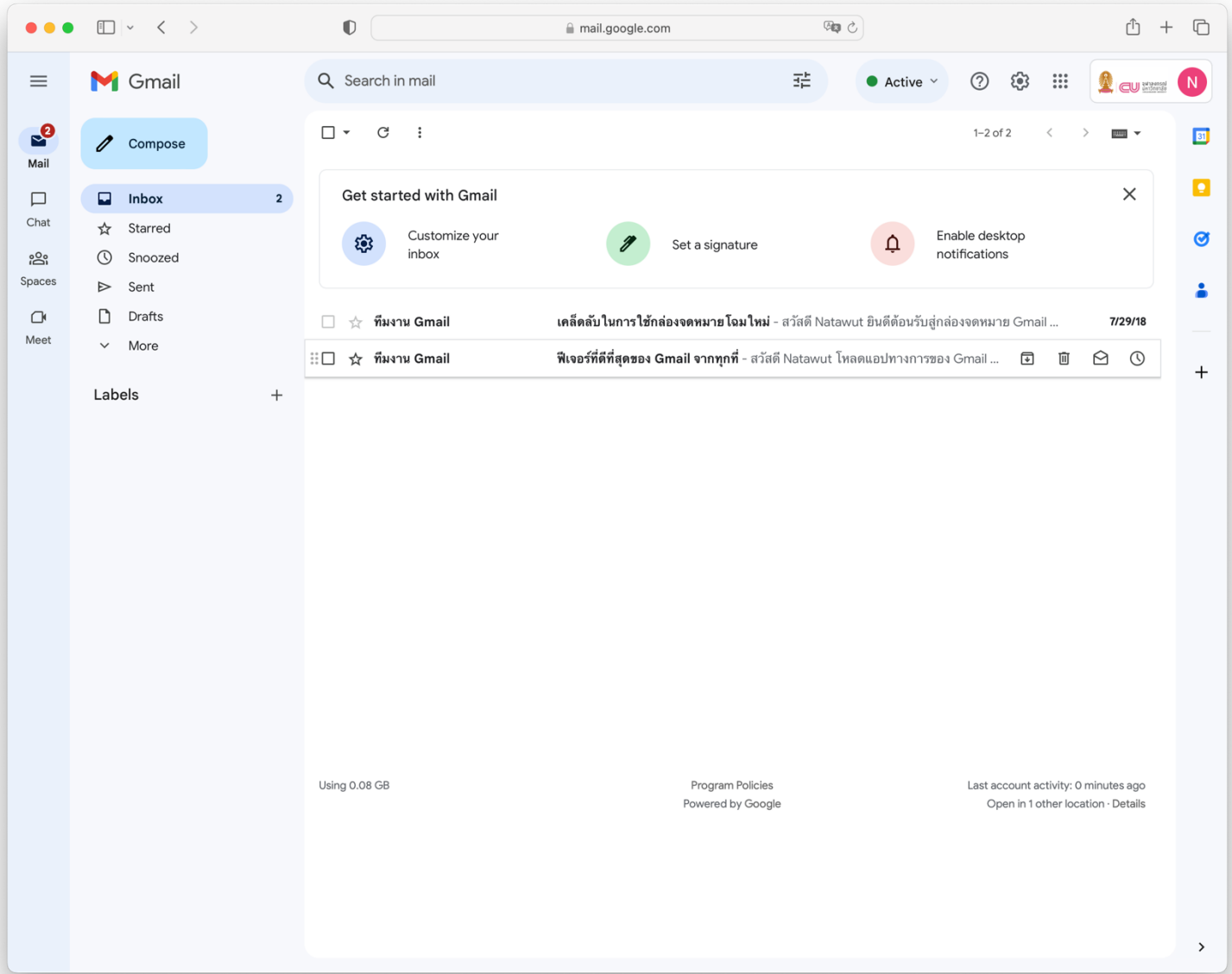
Data Extraction Steps

- Data Loading
  - Save to database
  - Save to data file
  - Save to dataframe

Scraping a Web Page

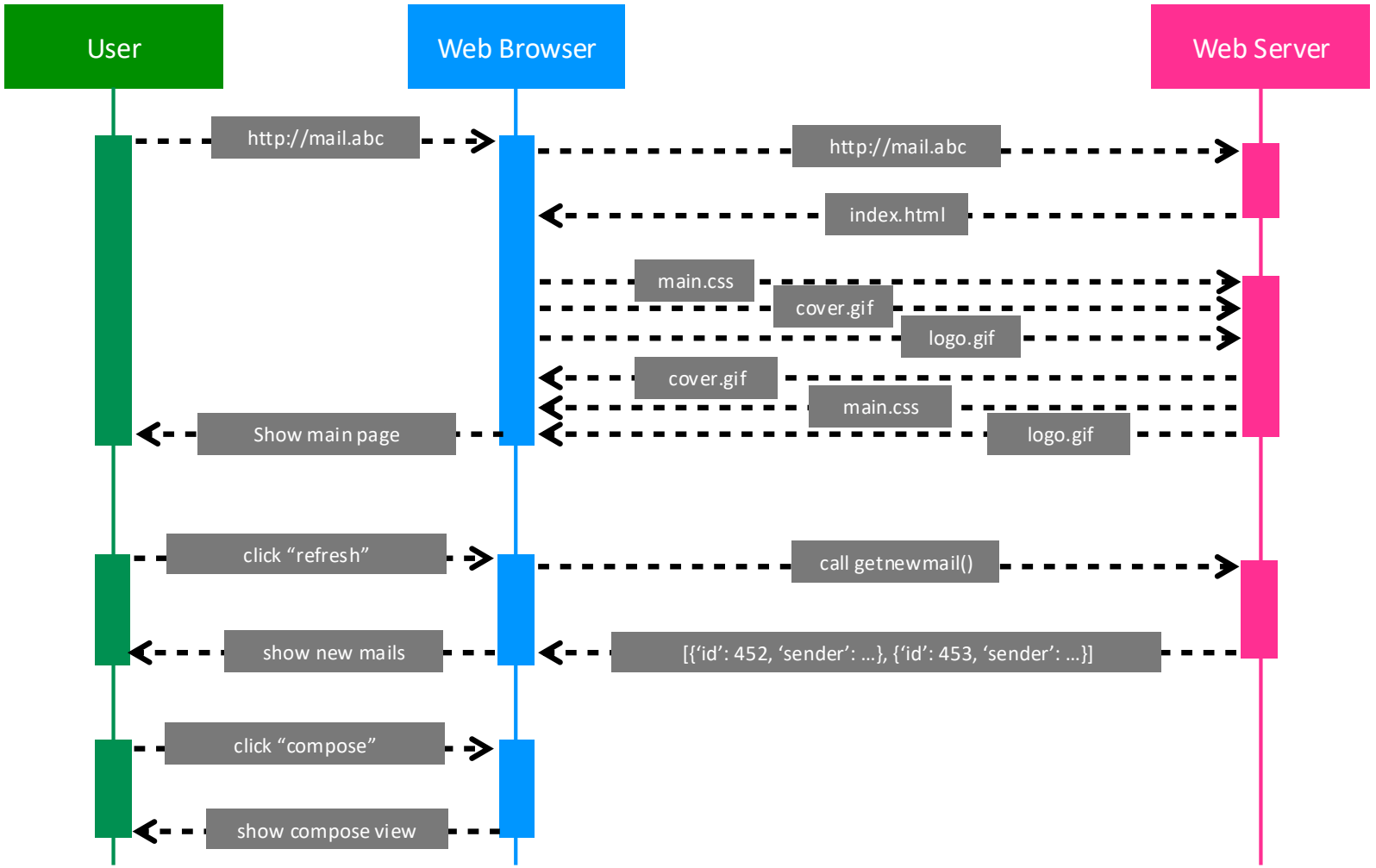
Understand HTML and how HTTP works





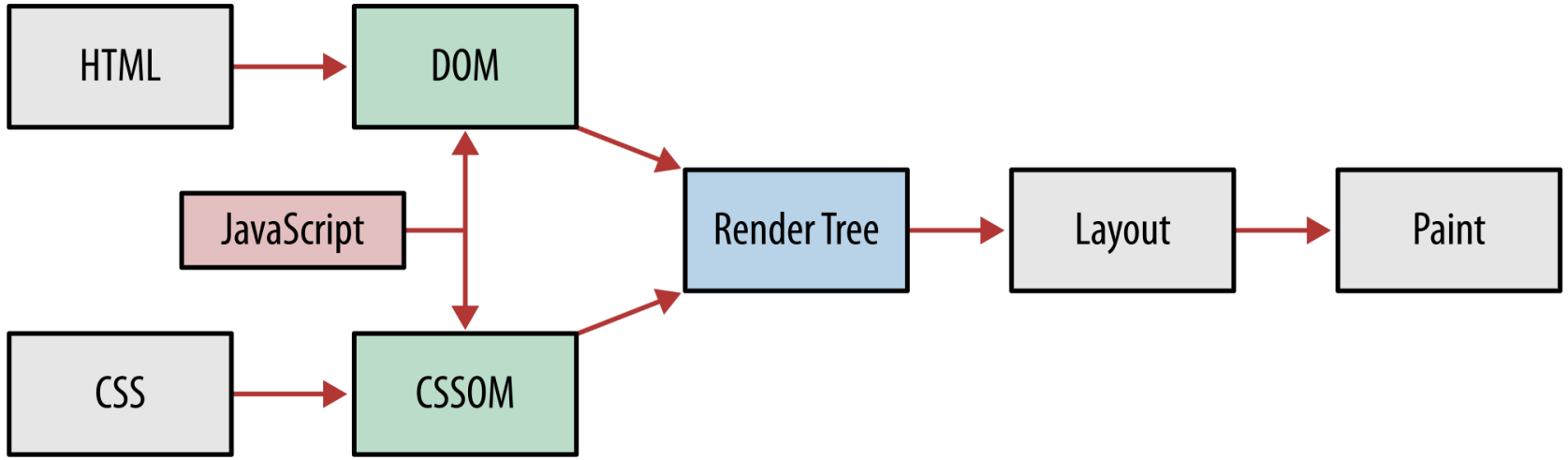
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Understand Web Page Mechanism



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Understand Web Page Mechanism



Source: https://hpbn.co/primer-on-web-performance/

Run ›

Result Size: 426 x 495

Get your own website

```
<!DOCTYPE html>
<html>
<body>

<p>Click the button to display the domain name of the server
that loaded this document.</p>

<button onclick="myFunction()">Try it</button>

<p id="demo"></p>

<script>
function myFunction() {
  document.getElementById("demo").innerHTML =
document.domain;
}
</script>

</body>
</html>
```

Click the button to display the domain name of the server that loaded this document.

Try it

www.w3schools.com

DOM manipulation

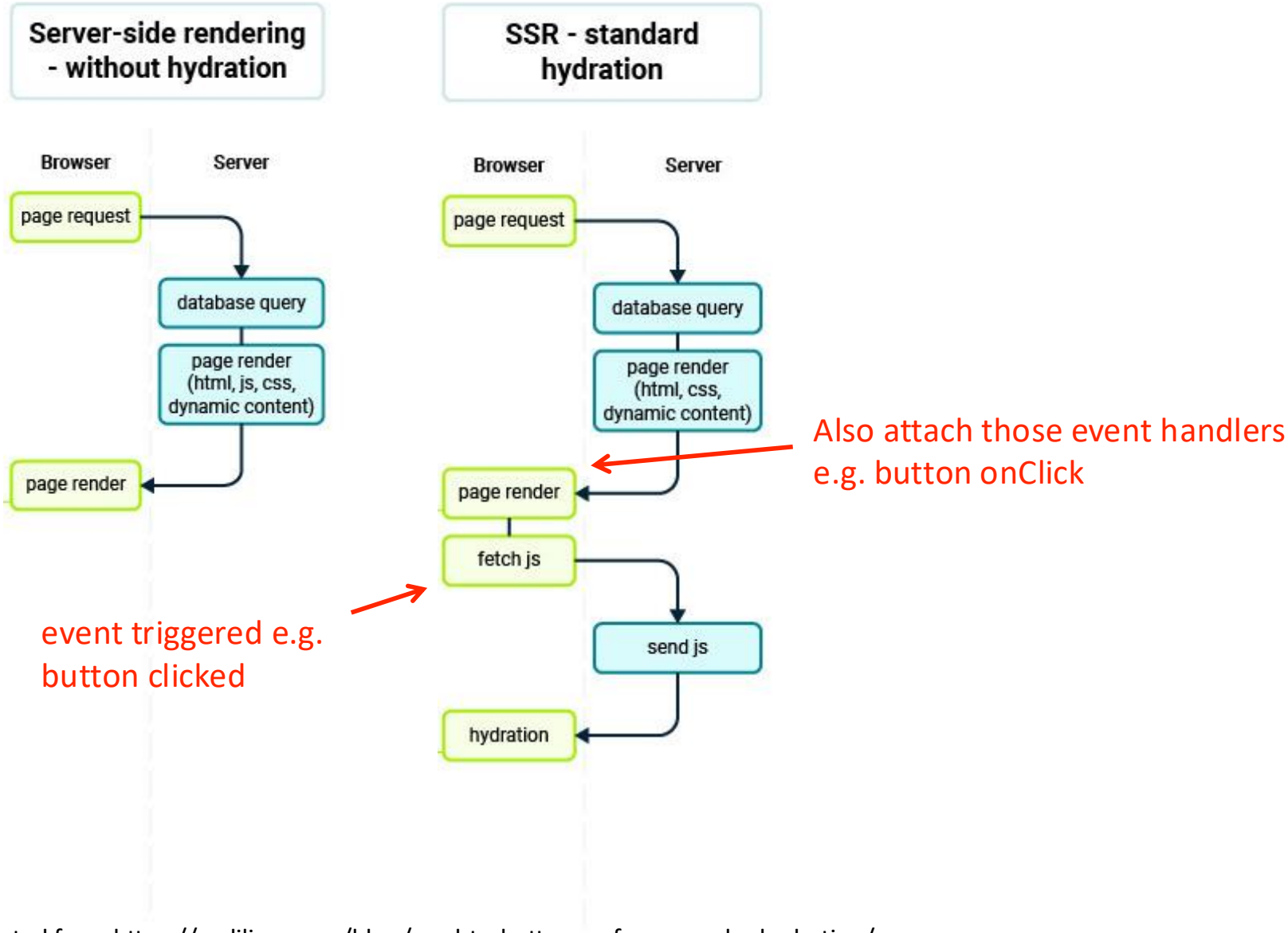
https://www.w3schools.com/js/tryit.asp?filename=tryjs_doc_domain

Understand Web Rendering Strategies

- Rendering = combining templates and data to create web content
- Server-Side Rendering (SSR)
  - Server generates a complete HTML page for every request
  - Ideal for sites needing fresh, real-time data while maintaining high SEO performance
  - Support by most web frameworks, including modern ones
- Client-Side Rendering (CSR)
  - Browser receives a near-empty HTML file and uses JavaScript to render content. It provides a smooth, app-like experience for user-specific, highly interactive interfaces
  - Support by all modern web frameworks e.g. React, Angular, Next.js, SvelteKit

Understand Web Rendering Strategies

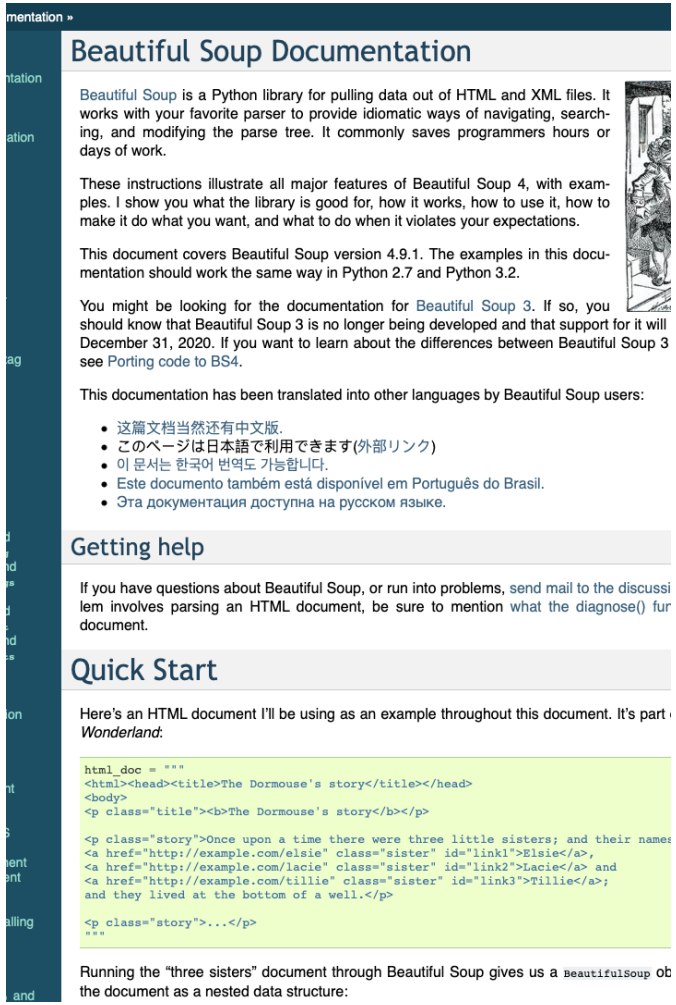
- Static-Site Generation (SSG)
  - Pages are pre-rendered into static HTML files at build time
  - SEO friendly, extremely fast load times and is best for blogs, marketing sites, or documentation
  - Support by some modern frameworks e.g. Astro, Gatsby, Next.js, SvelteKit
- SSR with Hydration
  - Server-side Rendering and the event handlers are added on the client-side
  - Support by some modern frameworks e.g. Astro, Next.js, SvelteKit



Adapted from https://codilime.com/blog/road-to-better-performance-by-hydration/

Parsing A Simple Webpage

- Applicable for a simple page only
  - No content rendered by javascript
  - Parse HTML with BeautifulSoup
- BeautifulSoup allows
  - Parsing HTML
  - Accessing and Navigating the DOM tree
  - Getting information from nodes



mentation »

Beautiful Soup Documentation

Beautiful Soup is a Python library for pulling data out of HTML and XML files. It works with your favorite parser to provide idiomatic ways of navigating, searching, and modifying the parse tree. It commonly saves programmers hours or days of work.

These instructions illustrate all major features of Beautiful Soup 4, with examples. I show you what the library is good for, how it works, how to use it, how to make it do what you want, and what to do when it violates your expectations.

This document covers Beautiful Soup version 4.9.1. The examples in this documentation should work the same way in Python 2.7 and Python 3.2.

You might be looking for the documentation for Beautiful Soup 3. If so, you should know that Beautiful Soup 3 is no longer being developed and that support for it will December 31, 2020. If you want to learn about the differences between Beautiful Soup 3 see Porting code to BS4.

This documentation has been translated into other languages by Beautiful Soup users:

- 这篇文档当然还有中文版.
- このページは日本語で利用できます(外部リンク)
- 이 문서는 한국어 번역도 가능합니다.
- Este documento também está disponível em Português do Brasil.
- Эта документация доступна на русском языке.

Getting help

If you have questions about Beautiful Soup, or run into problems, send mail to the discussi lem involves parsing an HTML document, be sure to mention what the diagnose() fur document.

Quick Start

Here's an HTML document I'll be using as an example throughout this document. It's part *Wonderland*:

```
html_doc = """
<html><head><title>The Dormouse's story</title></head>
<body>
<p class="title"><b>The Dormouse's story</b></p>

<p class="story">Once upon a time there were three little sisters; and their names
<a href="http://example.com/elsie" class="sister" id="link1">Elsie</a>,
<a href="http://example.com/lacie" class="sister" id="link2">Lacie</a> and
<a href="http://example.com/tillie" class="sister" id="link3">Tillie</a>;
and they lived at the bottom of a well.</p>

<p class="story">...</p>
"""
```

Running the "three sisters" document through Beautiful Soup gives us a `BeautifulSoup` ob the document as a nested data structure:

https://www.w3schools.com/howto/tryit.asp?filename=tryhow_css_example_website

```html
<!DOCTYPE html>
<html lang="en">
<head>
<title>Page Title</title>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<style>
* {
  box-sizing: border-box;
}

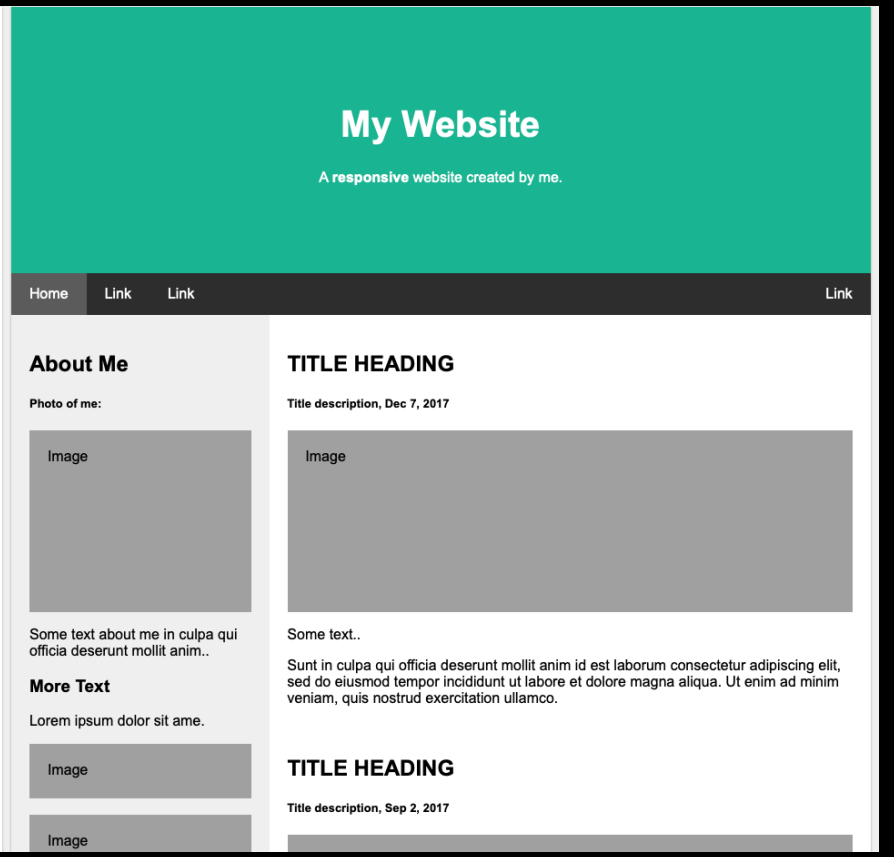
/* Style the body */
body {
  font-family: Arial, Helvetica, sans-serif;
  margin: 0;
}

/* Header/logo Title */
.header {
  padding: 80px;
  text-align: center;
  background: #1abc9c;
  color: white;
}

/* Increase the font size of the heading */
.header h1 {
  font-size: 40px;
}

/* Sticky navbar - toggles between relative and fixed, depending on the scroll position. It is
positioned relative until a given offset position is met in the viewport - then it "sticks" in place
(like position:fixed). The sticky value is not supported in IE or Edge 15 and earlier versions.
However, for these versions the navbar will inherit default position */
.navbar {
  overflow: hidden;
  background-color: #333;
  position: sticky;
  position: -webkit-sticky;
  top: 0;
}

/* Style the navigation bar links */
.navbar a {
  float: left;
  display: block;
  color: white;
  text-align: center;
  padding: 14px 20px;
  text-decoration: none;
}
```



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simple_page.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
<title>Page Title</title>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<style>
* {
  box-sizing: border-box;
}

/* Style the body */
body {
  font-family: Arial, Helvetica, sans-serif;
  margin: 0;
}

/* Header/logo Title */
.header {
  padding: 80px;
  text-align: center;
  background: #1abc9c;
  color: white;
}

/* Increase the font size of the heading */
.header h1 {
  font-size: 40px;
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/* Sticky navbar - toggles between relative and fixed, depending on the scroll position. It is positioned relative until
a given offset position is met in the viewport - then it "sticks" in place (like position:fixed). The sticky value is
not supported in IE or Edge 15 and earlier versions. However, for these versions the navbar will inherit default
position */
.navbar {
  overflow: hidden;
  background-color: #333;
  position: sticky;
  position: -webkit-sticky;
  top: 0;
}
```

```
<body>

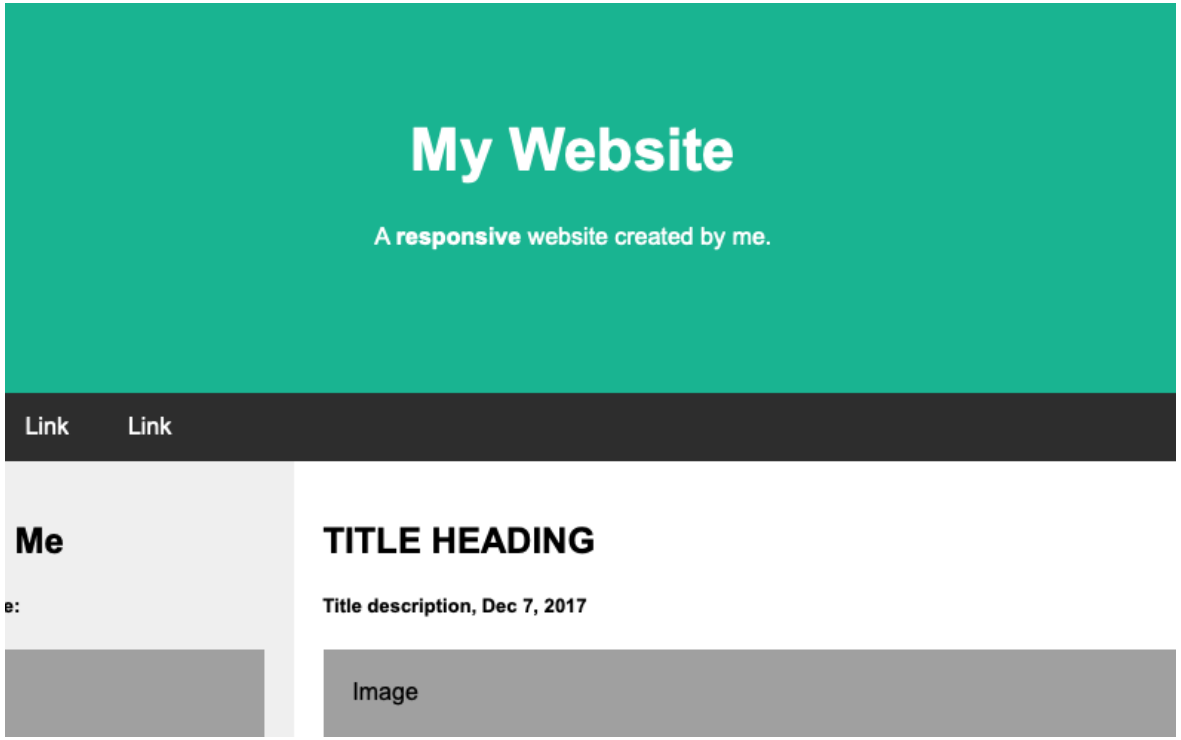
<div class="header">
  <h1>My Website</h1>
  <p>A <b>responsive</b> website created by me.</p>
</div>

<div class="navbar">
  <a href="#" class="active">Home</a>
  <a href="#">Link</a>
  <a href="#">Link</a>
  <a href="#" class="right">Link</a>
</div>

<div class="row">
  <div class="side">
    <h2>About Me</h2>
    <h5>Photo of me:</h5>
    <div class="fakeimg" style="height:200px;">Image</div>
    <p>Some text about me in culpa qui officia deserunt mollit anim..</p>
    <h3>More Text</h3>
    <p>Lorem ipsum dolor sit ame.</p>
    <div class="fakeimg" style="height:60px;">Image</div><br>
    <div class="fakeimg" style="height:60px;">Image</div><br>
    <div class="fakeimg" style="height:60px;">Image</div>
  </div>
  <div class="main">
    <h2>TITLE HEADING</h2>
    <h5>Title description, Dec 7, 2017</h5>
    <div class="fakeimg" style="height:200px;">Image</div>
    <p>Some text..</p>
    <p>Sunt in culpa qui officia deserunt mollit anim id est laborum consectetur adipiscing elit, sed
do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud
exercitation ullamco.</p>
    <br>
    <h2>TITLE HEADING</h2>
    <h5>Title description, Sep 2, 2017</h5>
    <div class="fakeimg" style="height:200px;">Image</div>
    <p>Some text..</p>
    <p>Sunt in culpa qui officia deserunt mollit anim id est laborum consectetur adipiscing elit, sed
do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud
exercitation ullamco.</p>
  </div>
</div>

<div class="footer">
  <h2>Footer</h2>
</div>

</body>
```



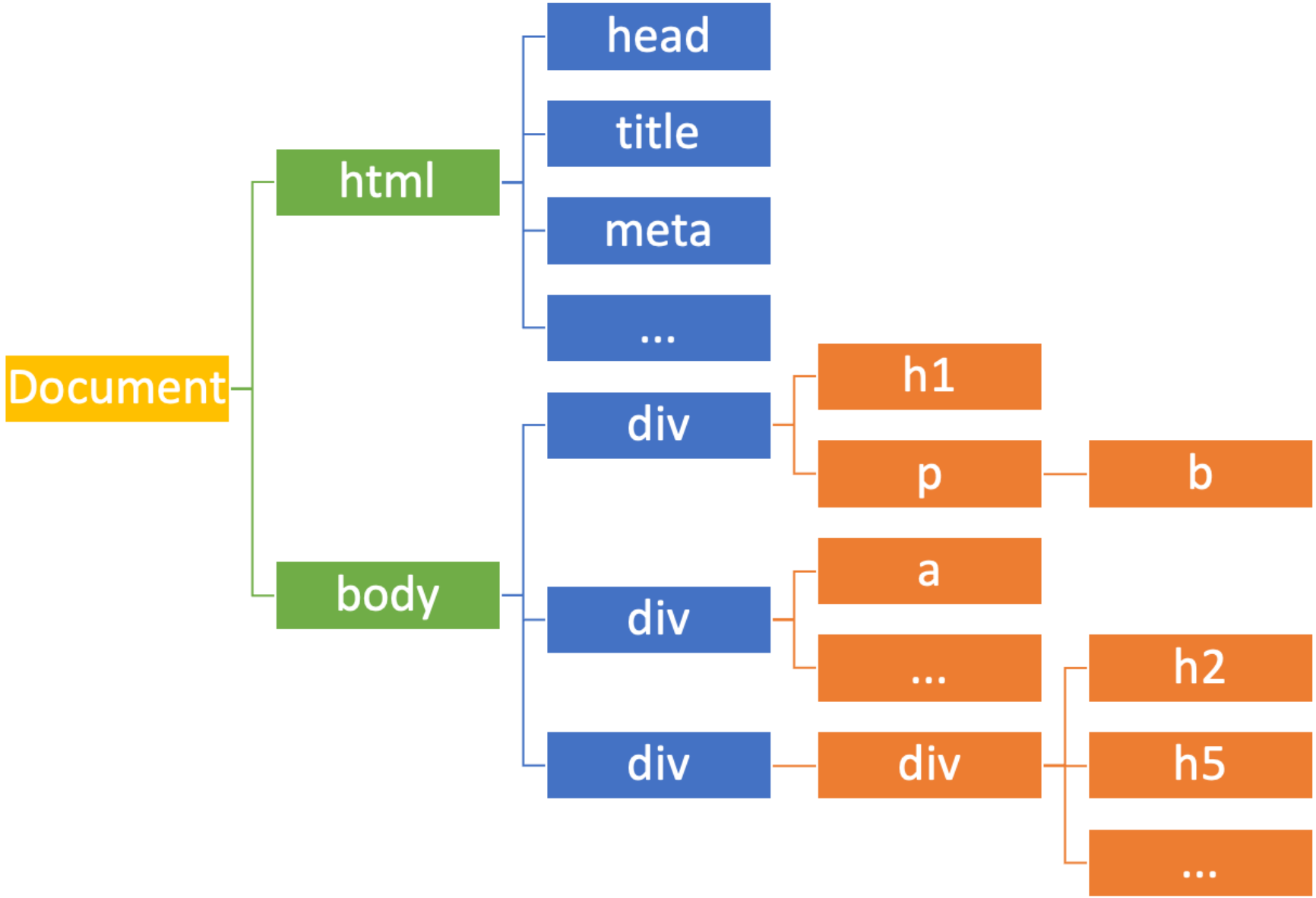
CSS

```css
/* Header/logo Title */
.header {
  padding: 80px;
  text-align: center;
  background: #1abc9c;
  color: white;
}

/* Increase the font size of the heading */
.header h1 {
  font-size: 40px;
}
```

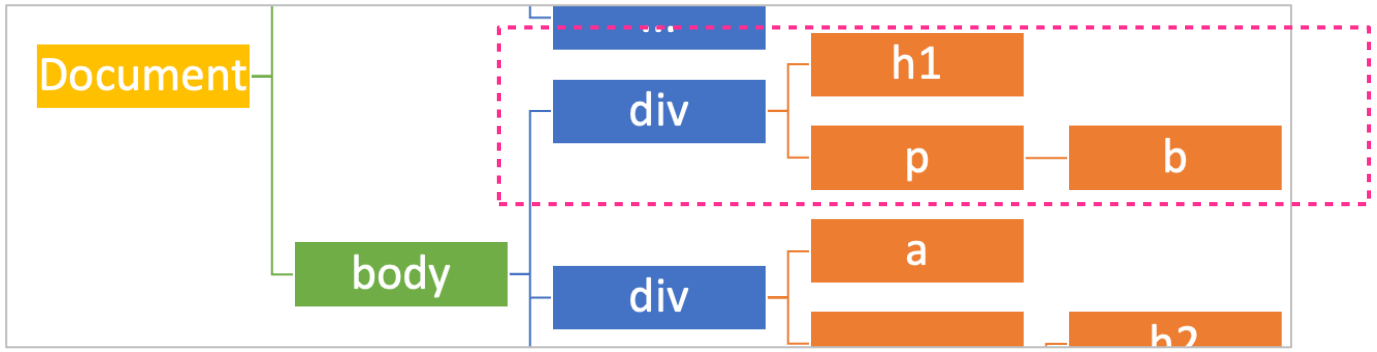
body

```html
<div class="header">
  <h1>My Website</h1>
  <p>A <b>responsive</b> website created by me.</p>
</div>
```



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DOM Node Example




```
<div class="header">
  <h1>My Website</h1>
  <p>A <b>responsive</b> website created by me.</p>
</div>
```

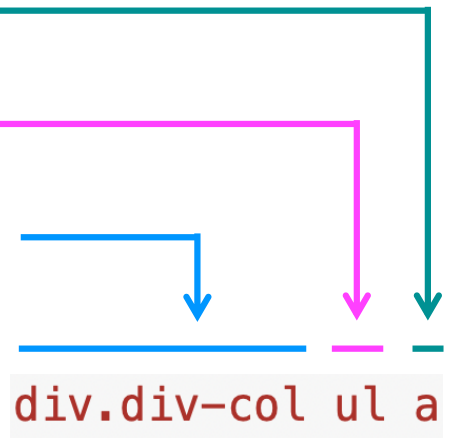
- Node ‘div’ has 2 children (h1 and p) and 1 attribute (key=‘class’, value=‘header’)

CSS Selector

- CSS Selector is a pattern of elements and other terms that tell the browser which HTML elements should be selected to have the CSS property values inside the rule applied to them
  - Elements with a specific tag e.g. all H2 elements
  - Elements with a specific class / id / attribute e.g. all elements with class = “fakeimg”
  - Combination of selectors e.g.
    - all div elements with class = “fakeimg”
    - all div elements or elements with class = “fakeimg”
- A selector may select multiple elements at once

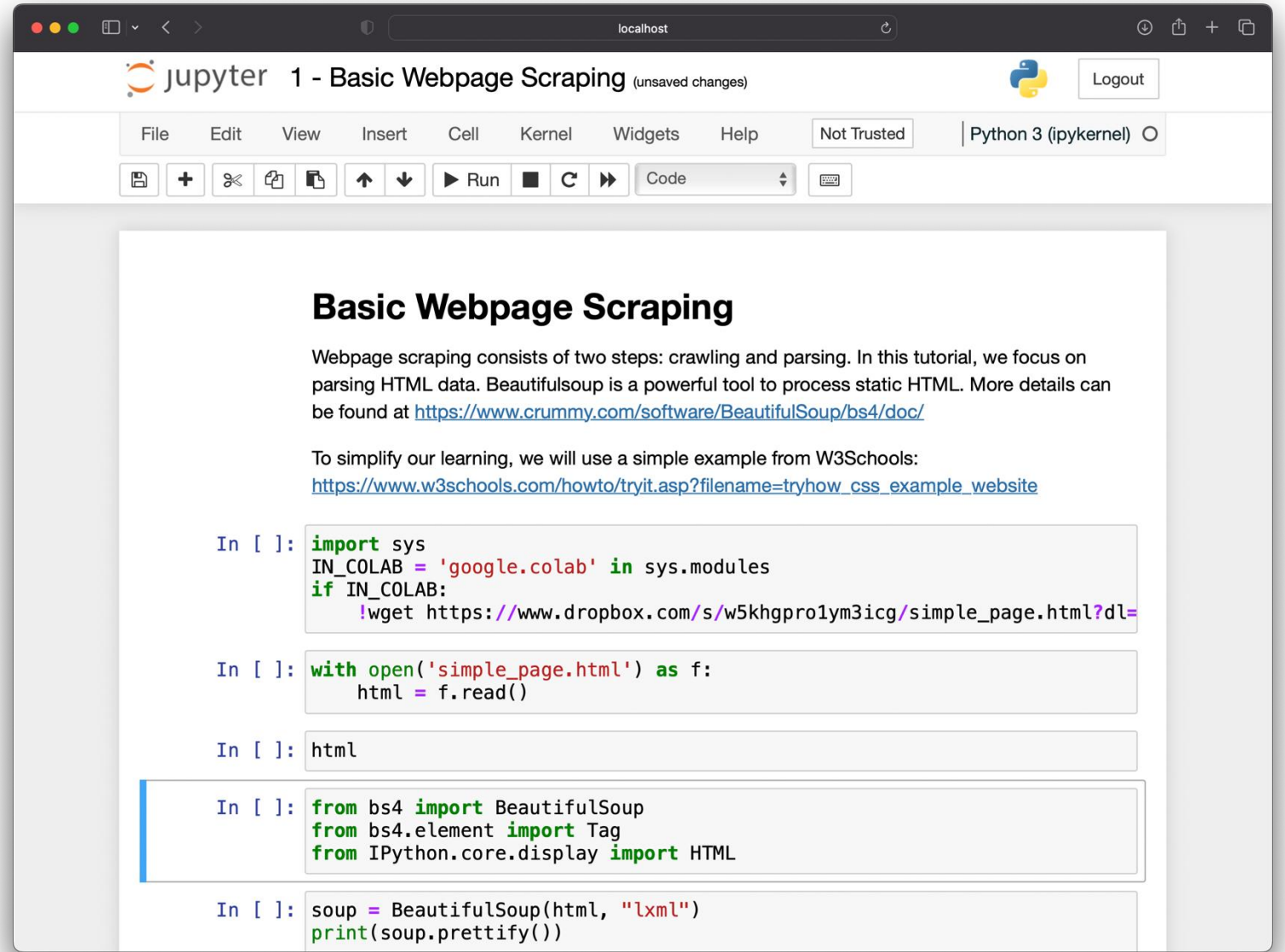
Understand CSS Selector Syntax

- CSS Selector is a powerful way to navigate DOM tree and locate DOM node(s)
- Find node(s) by tag, id, class, attributes, states (active, focus, hover, etc.), and much more
- Support hierarchical structure e.g. descendant, sibling, order of children, etc.
- In the example, we want to select
  - an anchor element (a) that is
  - inside an unordered list (ul), which is
  - inside a div element whose class is ‘div-col’



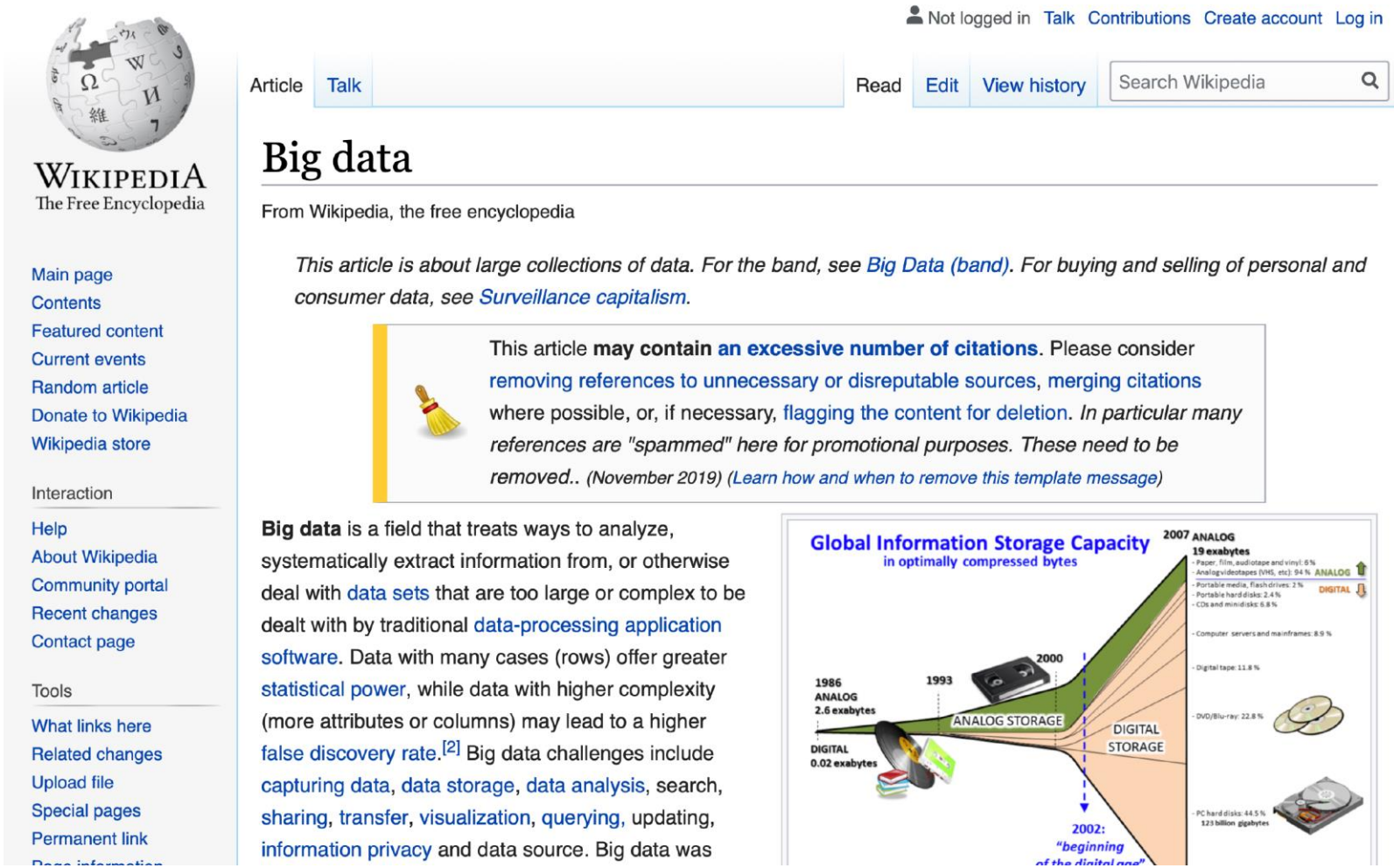
Selector	Example	Example description
.class	.intro	Selects all elements with class="intro"
.class1.class2	.name1.name2	Selects all elements with both *name1* and *name2* set within its class attribute
.class1 .class2	.name1 .name2	Selects all elements with *name2* that is a descendant of an element with *name1*
#id	#firstname	Selects the element with id="firstname"
*	*	Selects all elements
element	p	Selects all <p> elements
element.class	p.intro	Selects all <p> elements with class="intro"
element,element	div, p	Selects all elements and all <p> elements
element element	div p	Selects all <p> elements inside elements
element>element	div > p	Selects all <p> elements where the parent is a element
element+element	div + p	Selects the first <p> element that is placed immediately after elements
element1~element2	p ~ ul	Selects every element that is preceded by a <p> element
[*attribute*]	[target]	Selects all elements with a target attribute
[*attribute=value*]	[target=_blank]	Selects all elements with target="_blank"
[*attribute~=value*]	[title~=flower]	Selects all elements with a title attribute containing the word "flower"
[*attribute|=value*]	[lang|=en]	Selects all elements with a lang attribute value equal to "en" or starting with "en-"
[*attribute^=value*]	a[href^="https"]	Selects every <a> element whose href attribute value begins with "https"

https://www.w3schools.com/cssref/css_selectors.asp



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Example: Extracting references from a wiki page



WIKIPEDIA
The Free Encyclopedia

Not logged in Talk Contributions Create account Log in

Article Talk | Read Edit View history | Search Wikipedia

Main page
Contents
Featured content
Current events
Random article
Donate to Wikipedia
Wikipedia store

Interaction
Help
About Wikipedia
Community portal
Recent changes
Contact page

Tools
What links here
Related changes
Upload file
Special pages
Permanent link

Big data

From Wikipedia, the free encyclopedia

This article is about large collections of data. For the band, see Big Data (band). For buying and selling of personal and consumer data, see Surveillance capitalism.

This article **may contain an excessive number of citations**. Please consider removing references to unnecessary or disreputable sources, merging citations where possible, or, if necessary, flagging the content for deletion. *In particular many references are "spammed" here for promotional purposes. These need to be removed.*. *(November 2019) (Learn how and when to remove this template message)*

Big data is a field that treats ways to analyze, systematically extract information from, or otherwise deal with data sets that are too large or complex to be dealt with by traditional data-processing application software. Data with many cases (rows) offer greater statistical power, while data with higher complexity (more attributes or columns) may lead to a higher false discovery rate.[2] Big data challenges include capturing data, data storage, data analysis, search, sharing, transfer, visualization, querying, updating, information privacy and data source. Big data was

Contents [hide]

(Top)

Definition

Characteristics

Architecture

Technologies

Applications

Case studies

Research activities

Critique

See also

References

Further reading

External links

- Encouraging members of society to abandon interactions with institutions that would create a digital trace, thus creating obstacles to social inclusion

If these potential problems are not corrected or regulated, the effects of big data policing may continue to shape societal hierarchies. Conscientious usage of big data policing could prevent individual level biases from becoming institutional biases, Brayne also notes.

See also [edit]

For a list of companies, and tools, see also: Category:Big data

- Big data ethics – Ethics of mass data analytics
- Big data maturity model – Aspect of computer science
- Big memory – A large amount of random-access memory
- Data curation – work performed to ensure meaningful and enduring access to data
- Data defined storage – Marketing term for managing data by combining application, information and storage tiers
- Data engineering – Software engineering approach to designing and developing information systems
- Data lineage – Origins and events of data
- Data philanthropy – Aspect of culture
- Data science – Interdisciplinary field of study on deriving knowledge and insights from data
- Datafication – Technological trend
- Document-oriented database – Type of computer program
- List of big data companies
- Very large database – type of database containing a very large amount of data, so much that it can require specialized architectural, management, processing and maintenance methodologies
- XLDB – annual conference series on databases, data management and analytics

References [edit]

1. ^ Hilbert, Martin; López, Priscila (2011). "The World's Technological Capacity to Store, Communicate, and Compute Information". *Science*. **332** (6025): 60–65. Bibcode:2011Sci...332...60H. doi:10.1126/science.1200970. PMID 21310967. S2CID 206531385. Archived from the original on 14 April 2016. Retrieved 13 April 2016.
2. ^ Breur, Tom (July 2016). "Statistical Power Analysis and the contemporary "crisis" in social sciences". *Journal of Marketing Analytics*. London, England: Palgrave Macmillan. **4** (2–3): 61–65. doi:10.1057/s41270-016-0001-3. ISSN 2050-3318.

104. ^ Josh Rogin (2 August 2018). "Ethnic cleansing makes a comeback – in China". No. Washington Post. Archived from the original on 31 March 2019. Retrieved 4 August 2018. "Add to that the unprecedented security and surveillance state in Xinjiang, which includes all-encompassing monitoring based on identity cards, checkpoints, facial recognition and the collection of DNA from millions of individuals. The authorities feed all this data into an artificial-intelligence machine that rates people's loyalty to the Communist Party in order to control every aspect of their lives."

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Brayne also notes.

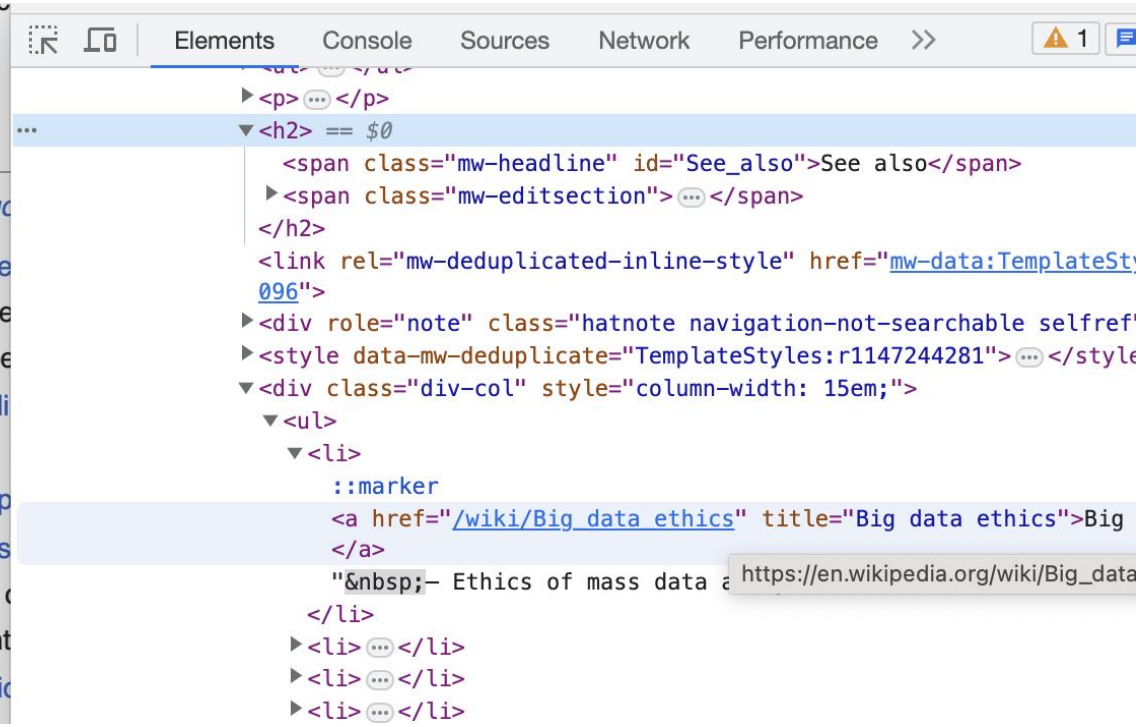
See also [edit]

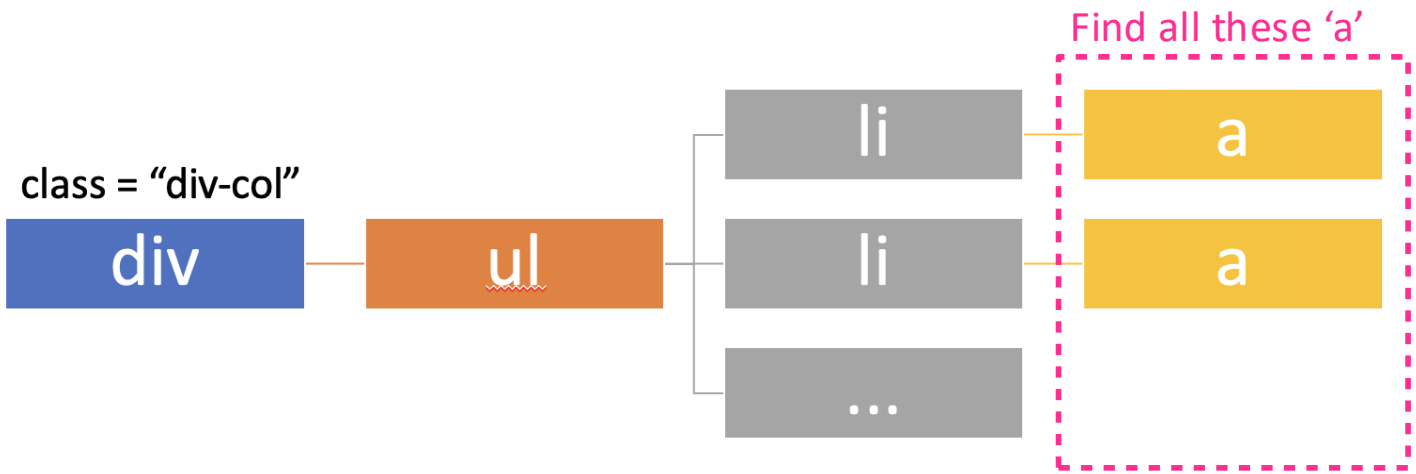
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panies, and tools, see also: Catego

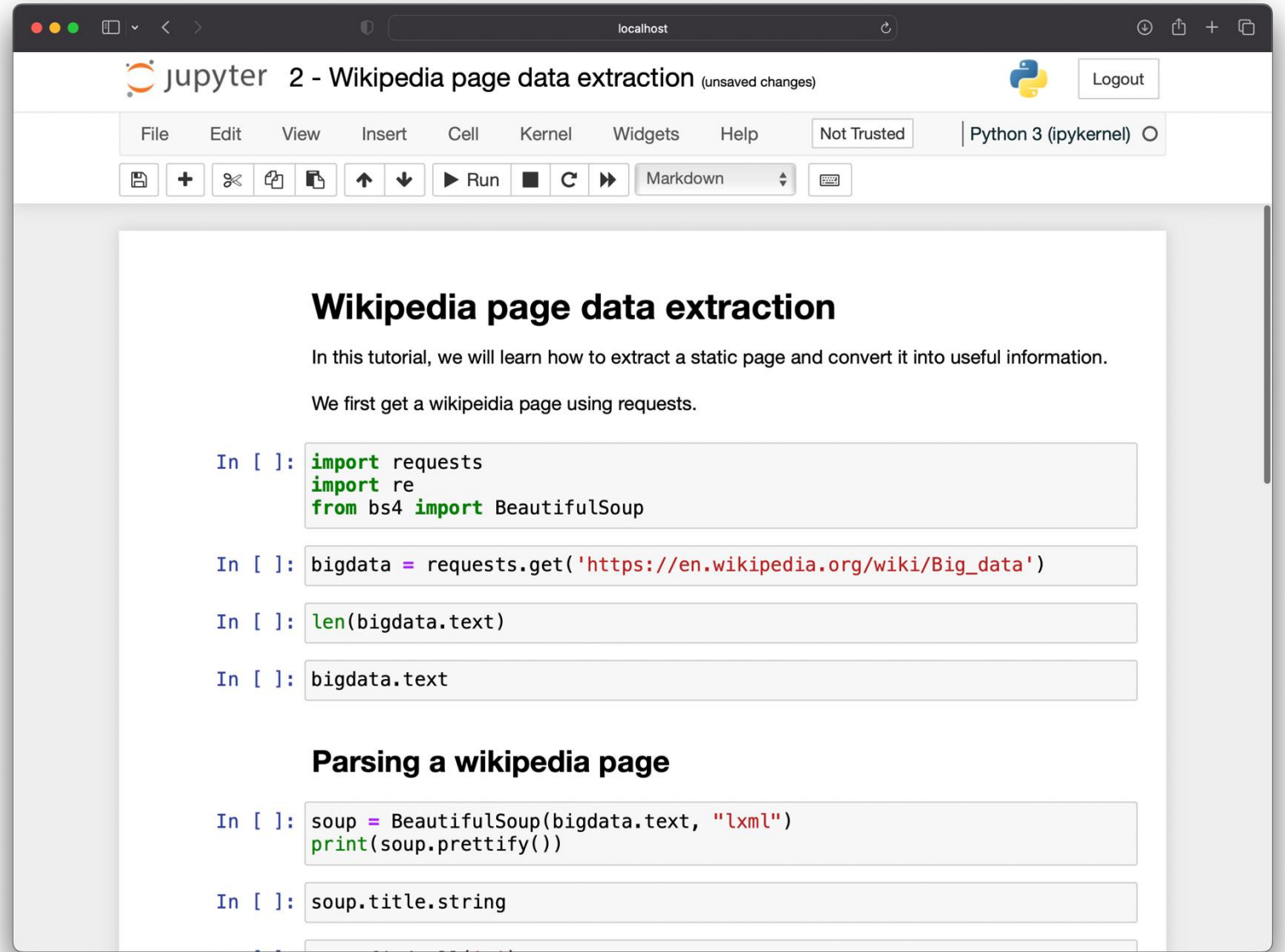
- Big data ethics – Ethics of mass data analytics
- Big data maturity model – Aspect of computer science
- Big memory – A large amount of random-access memory
- Data curation – work performed to ensure meaningful and enduring access to data
- Data defined storage – Marketing term

- Data e engine and de
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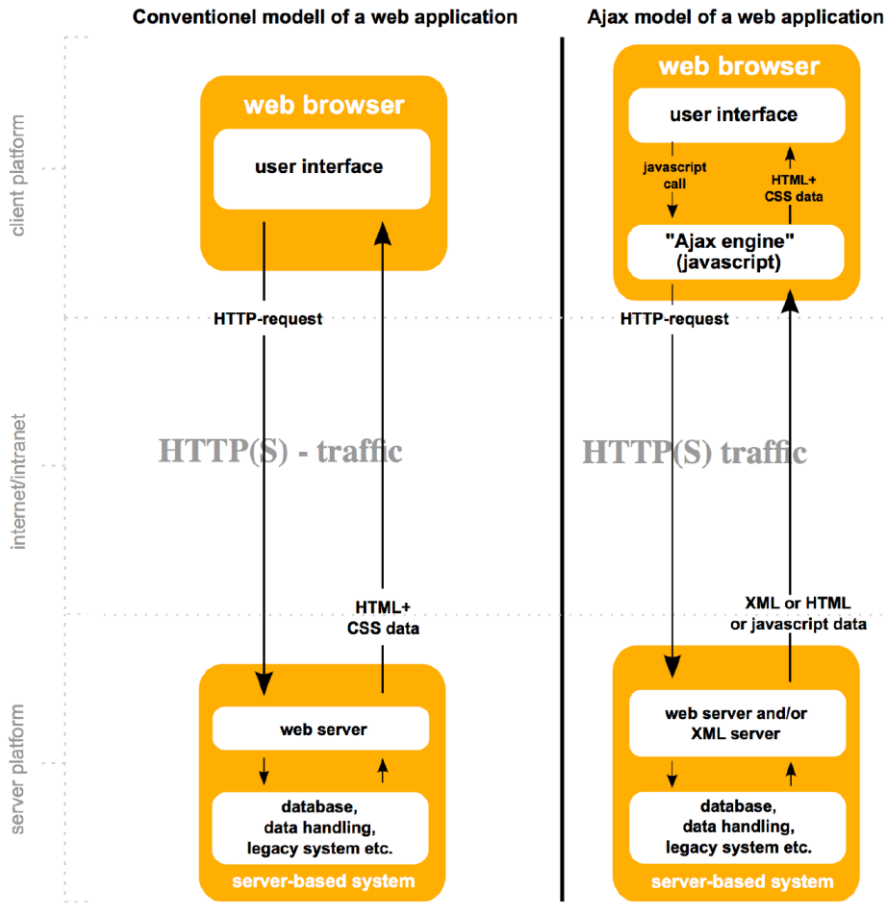
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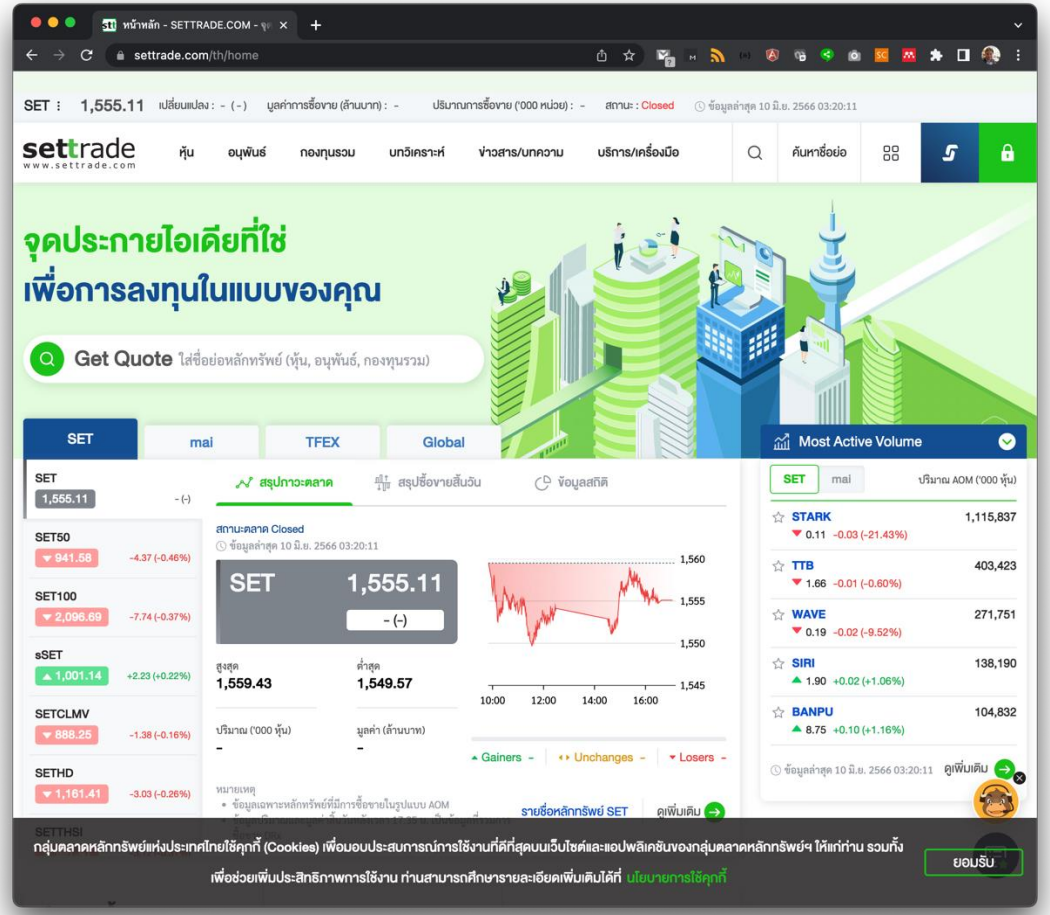
Extracting AJAX-based Webpage

- Modern web applications are AJAX-based e.g. SPA
- Typical web request approach does not work

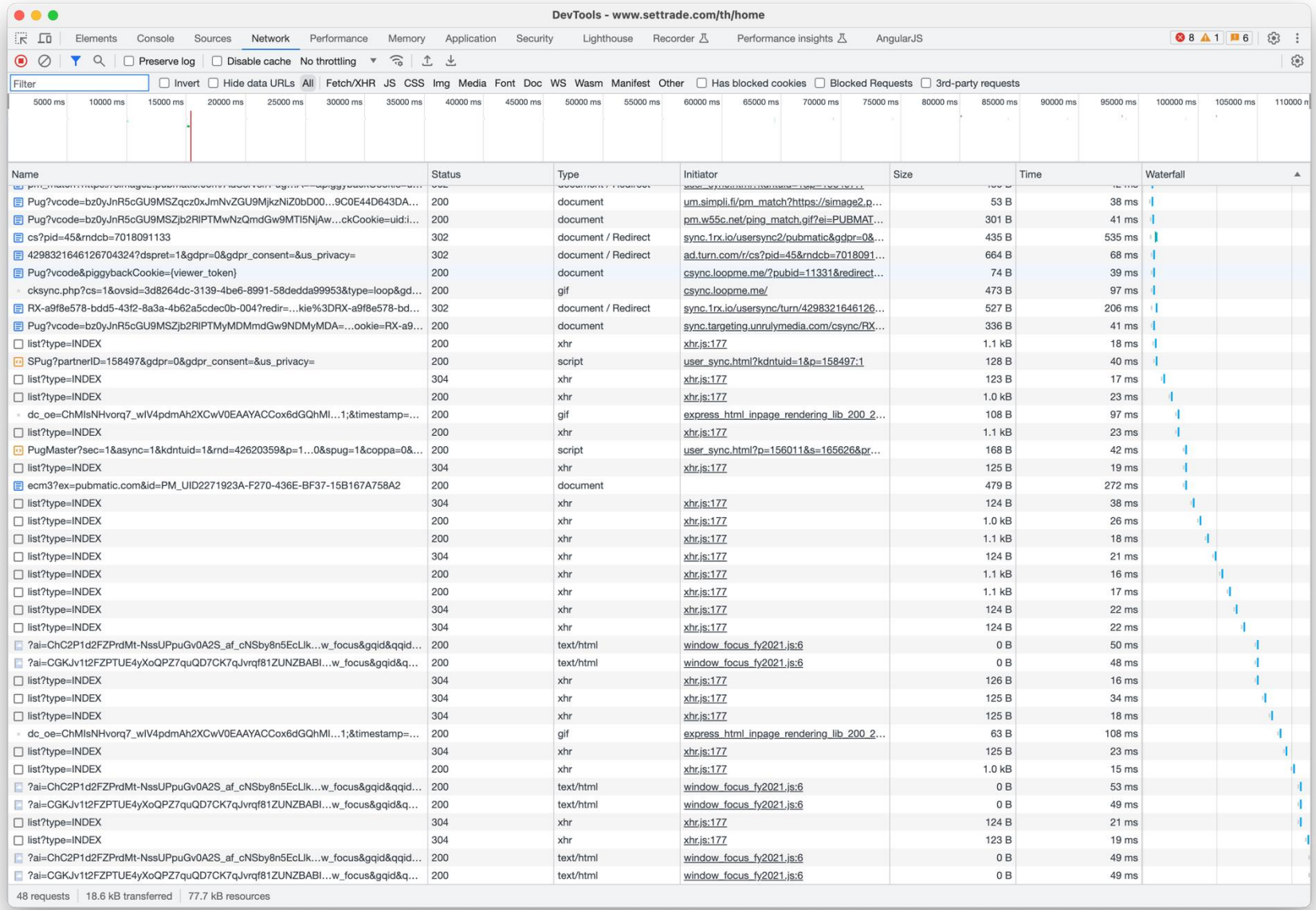


Source: https://en.wikipedia.org/wiki/Ajax_(programming)

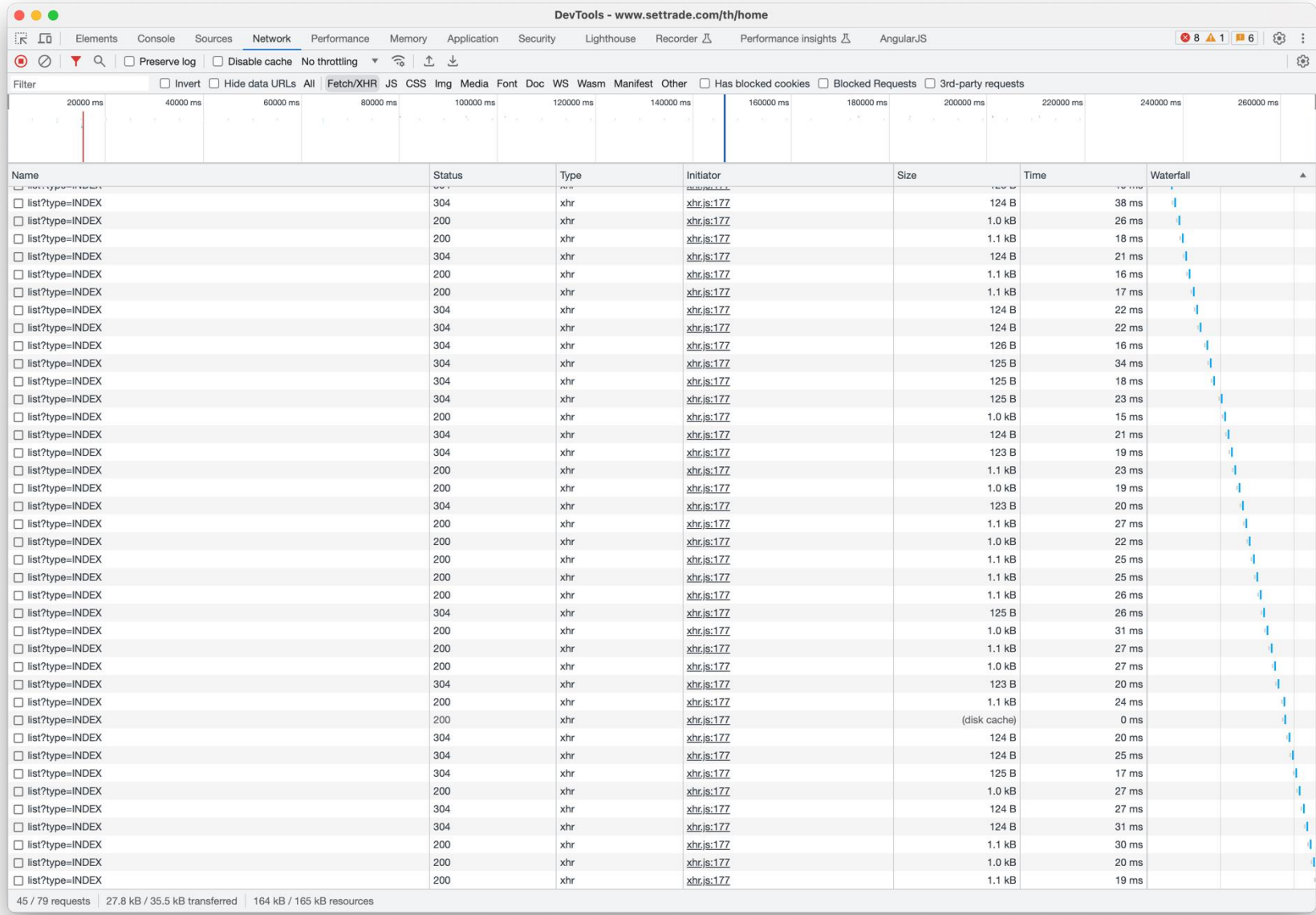
https://www.settrade.com/settrade/home



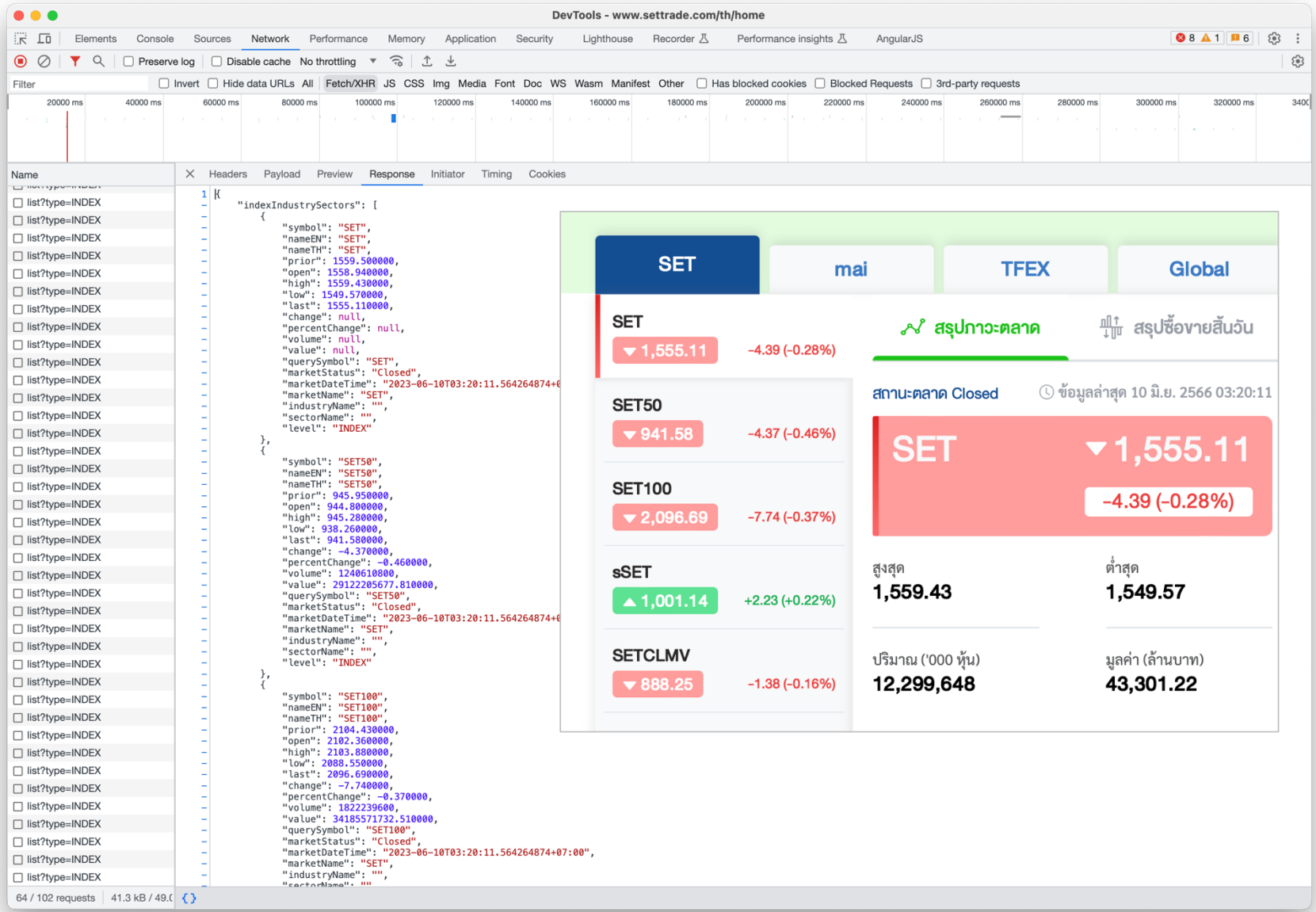
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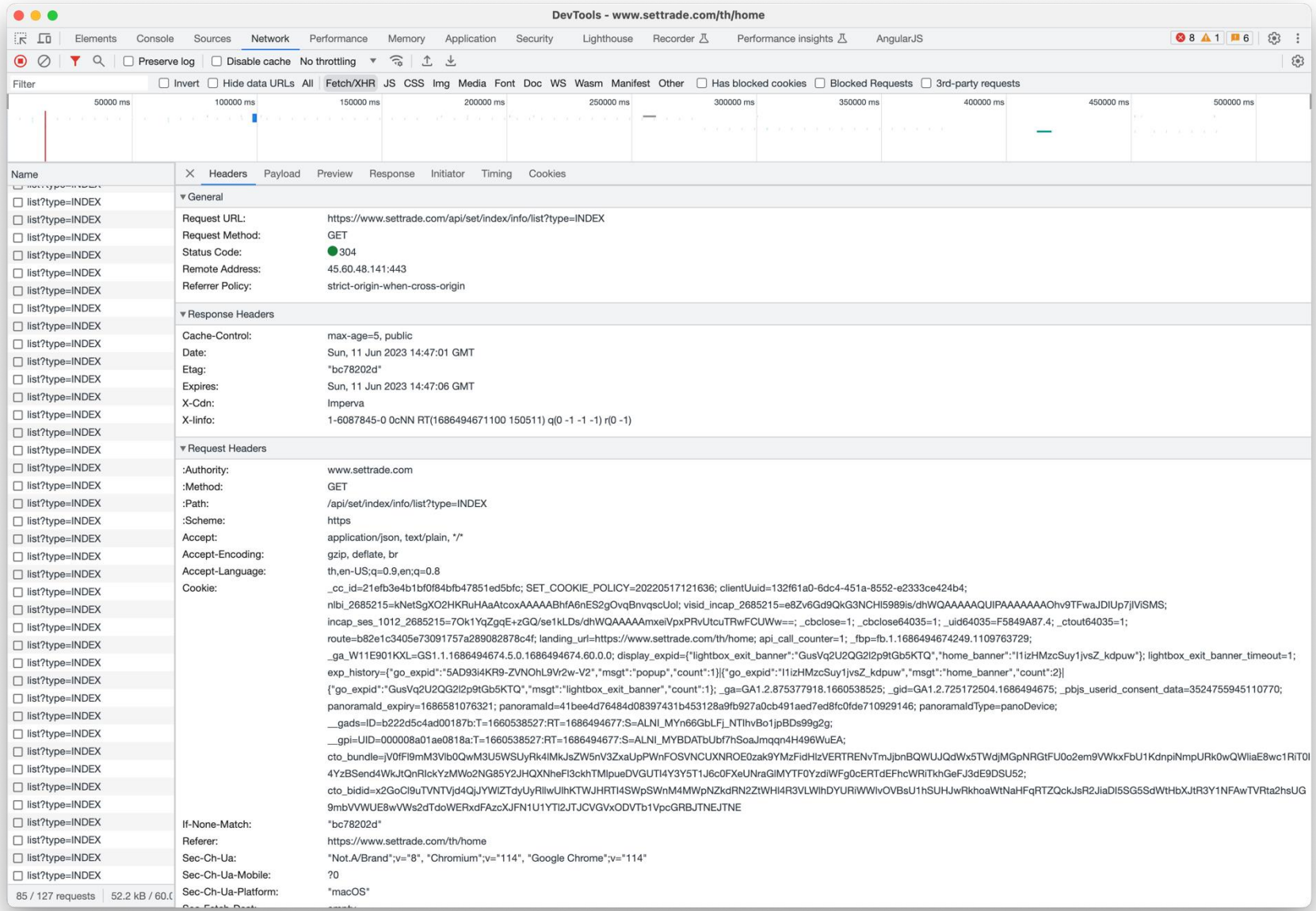
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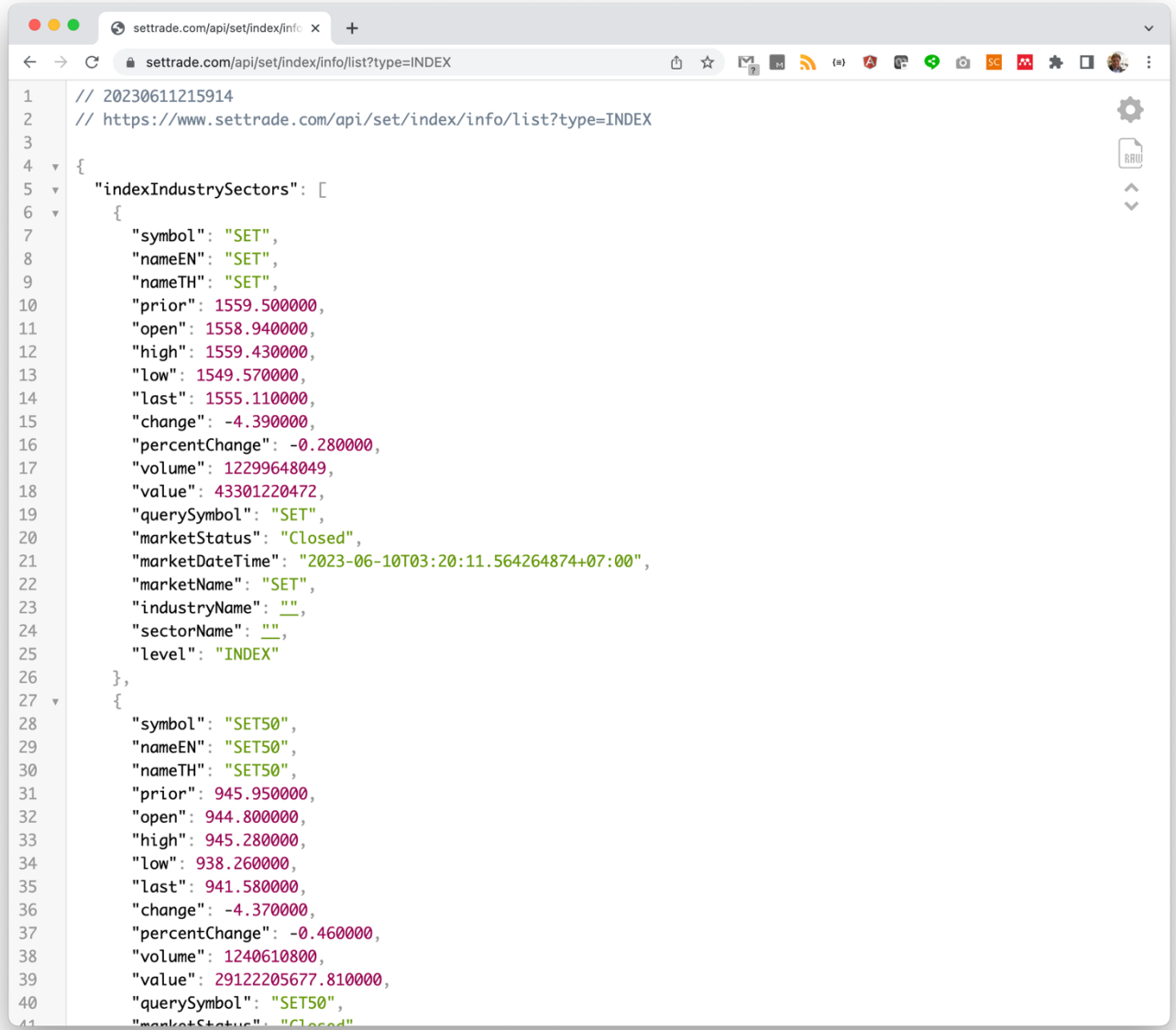
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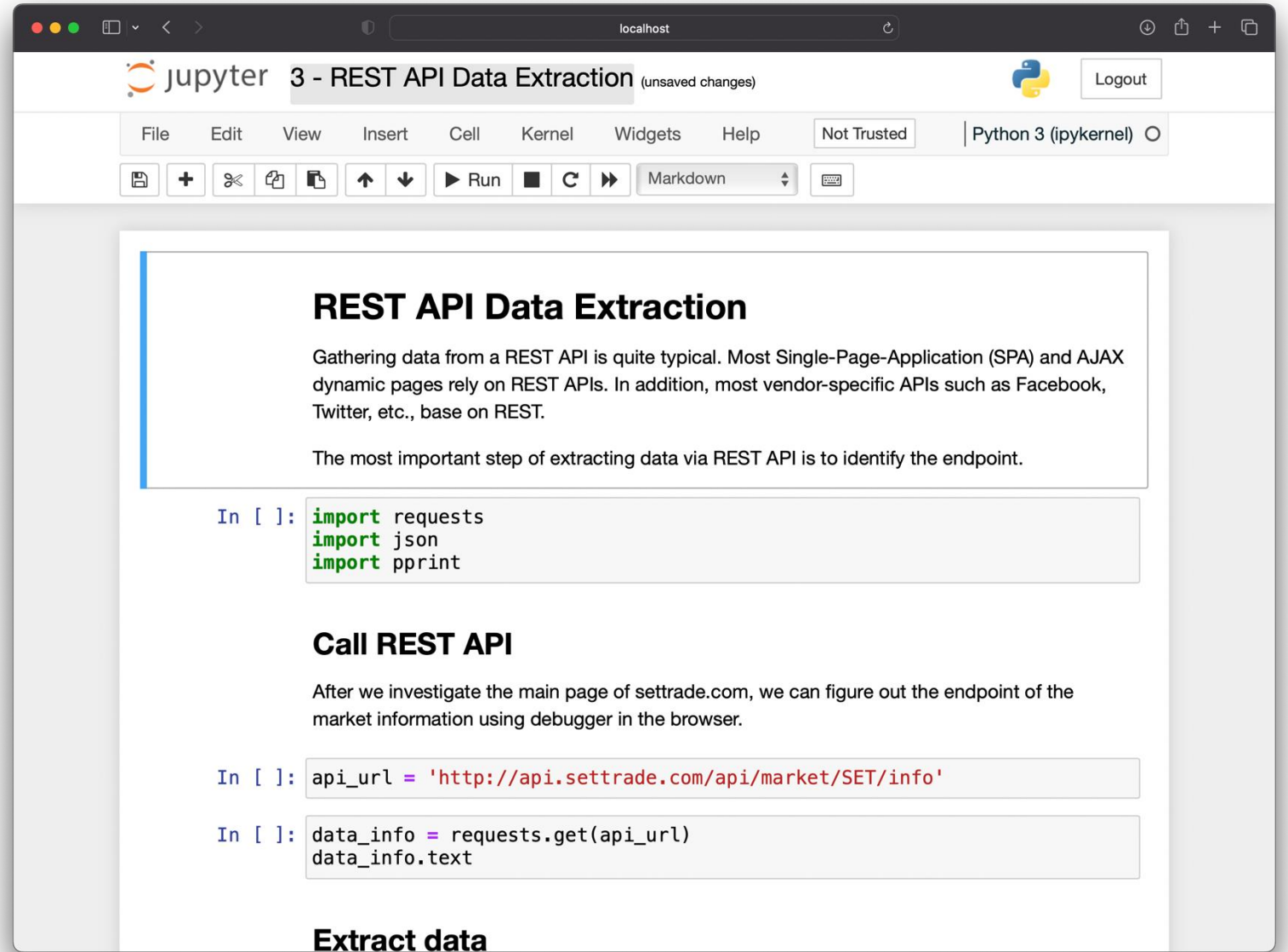
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Extract Complicated Webpage with Selenium

- Modern webpages are Javascript-based, which can be very difficult or impossible to use our previous techniques
  - Normal web applications usually base on session of multiple web pages using cookies and HTTP headers
  - Single Page Application (SPA) augments existing elements when being interacted and generates new elements on-the-fly
- Selenium is an open source framework that allow us to scrape data from all these web applications

Selenium Architecture



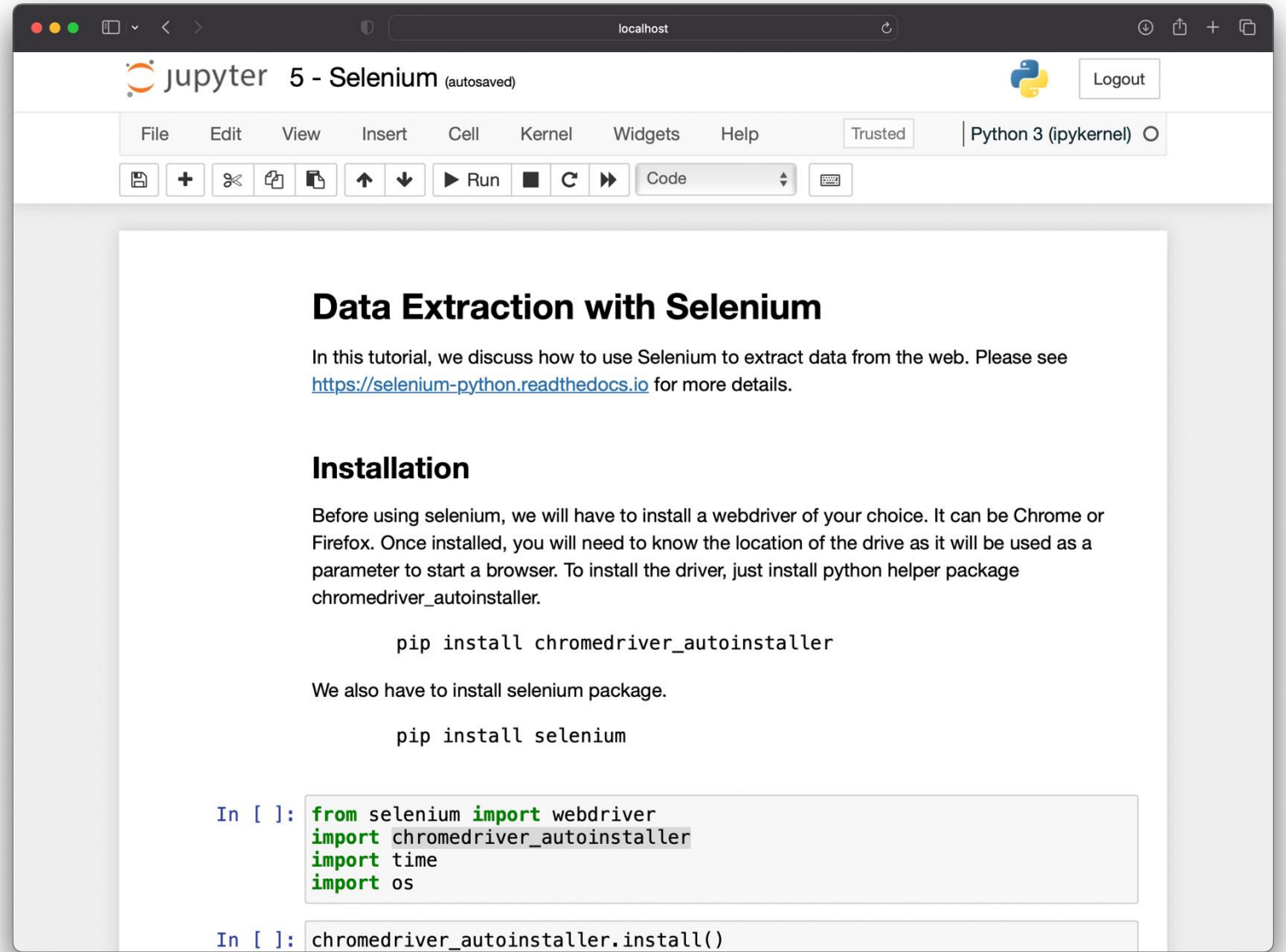
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Selenium Capabilities

- Process any website similar to normal web browser
  - Access a web page using normal URL
  - Grab HTML and other resource files in a particular web page
  - Find components based on criteria e.g. CSS selector, etc.
  - Interacting with components in a web page e.g. click, input keyboard
  - Navigate within a web page and across multiple pages
- During operation, Selenium can be normal mode or headless mode
- Checkout https://www.selenium.dev/documentation/ for more information

Example: Selenium on Google

- Visit https://www.google.com
- Get HTML of this web page
- Find a query box
- Enter ‘ประเทศไทย’ in the query box and press ‘enter’
- Search result links (‘a’ elements under an element with id=‘search’)
- Click the first link



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	Pros	Cons	Good For
BeautifulSoup	Fast	Only for non-interactive contents	SSR, SSG
API	Fast and Simple	For API only	CSR (API)
Selenium	Support complicated web sites	Slow	CSR, SSR with Hydration

Recommended Resources

- http://python.gotrained.com/python-json-api-tutorial/
- https://www.analyticsvidhya.com/blog/2015/10/beginner-guide-web-scraping-beautiful-soup-python/
- https://www.dataquest.io/blog/web-scraping-tutorial-python/
- https://www.datacamp.com/courses/importing-data-in-python-part-2
- https://code.tutsplus.com/tutorials/the-30-css-selectors-you-must-memorize--net-16048
- https://www.workingsoftware.dev/understanding-ssr-with-hydration-for-software-architects/